

Inference, Not Just Optimisation: Streaming Inversion-Free Quasi-Newton Estimation on Dependent Data Streams

Abstract

Streaming quasi-Newton methods are asymptotically efficient at first-order cost, but on a serially dependent stream their stated uncertainty is wrong by a factor that never shrinks. We show that one structural change fixes the estimate and the interval together, inside the same $O(dN)$ budget: consume the stream in *contiguous blocks* and *gap* the curvature updates, and the block gradients the algorithm already forms are exactly the ingredients of a consistent estimator of the long-run score covariance. The statistically correct object and the computational shortcut turn out to be the same operation, and the valid interval ends up cheaper than the invalid one it replaces.

The problem being fixed is this. Streaming quasi-Newton methods maintain an inverse-curvature matrix by rank-one updates, so a single pass costs $O(dN)$ for d parameters and N observations, the same order as stochastic gradient descent. Their theory assumes independent observations. Under dependence the limiting covariance is the *long-run* sandwich $H^{-1}SH^{-1}$, with S summing the score autocovariances at every lag, not the instantaneous $H^{-1}\Gamma_0H^{-1}$ that the independent-data theory certifies; nominal 95% intervals therefore converge to a coverage below 95% and stay there. On one of the real streams we study the estimated inflation is about 20, which pins that limit near a third.

We prove almost sure convergence, an $O(\log N/N)$ rate, and a central limit theorem with the long-run sandwich for the inversion-free rank-one curvature recursion under geometric mixing—the regime where the martingale argument behind the independent-data analysis fails. Because the blocks carry *gradients* rather than *iterates*, their length is set by the mixing time of the data rather than of the optimisation path: $b \asymp \log N$ suffices and S is estimated at $\tilde{O}(N^{-1/2})$, against $N^{-1/8}$ for the best published iterate-based estimators under dependence and $N^{-1/4}$ for that class’s minimax ceiling, with the reported sandwich converging at N^{-q} for any $q < 1/2$. We also give the exact asymptotic coverage of dependence-blind intervals, $2\Phi(z_{\alpha/2}/\sqrt{\kappa}) - 1$ in one variance-inflation factor κ , and verify it with no fitted constants. On autoregressive, Markov-modulated and dependent-logistic designs with κ between [TBD] and [TBD], and on hourly traffic and air-quality streams, our nominal 95% intervals cover [TBD]–[TBD] where dependence-blind intervals cover [TBD]–[TBD], at [TBD]–[TBD] times the width.

1 Introduction

Many estimation problems now arrive as a stream: observations appear one at a time or in small batches, they are too numerous to store, and a decision or a report is required at any moment. Stochastic approximation [74] is the natural tool, and averaged stochastic gradient descent [71, 78] is asymptotically efficient with $O(d)$ work per observation. Its weakness is conditioning: when the curvature of the objective has eigenvalues spread over orders of magnitude, first-order methods crawl [8, 51]. Second-order methods fix the conditioning but appear to cost $O(d^3)$ per step.

A line of work on *inversion-free* stochastic Newton methods removes that obstacle. When the Hessian has the form $\nabla^2 F(\theta) = \mathbb{E}[\alpha(\theta; \xi) \Psi(\theta; \xi) \Psi(\theta; \xi)^\top]$ —as it does for linear, logistic, softmax, Poisson and

ridge regression, and for the geometric median—the running curvature estimate is a sum of rank-one terms, so its inverse can be maintained directly by the Sherman–Morrison identity [80] without ever forming or inverting a matrix [6, 9, 12, 32]. Godichon-Baggioni and Werge [33] push this to its natural conclusion: thinning the rank-one updates and taking mini-batches of size d gives an estimator whose *total* cost is $O(dN)$ —first-order cost—while remaining asymptotically efficient. A concurrent construction achieves the same budget by masking Hessian columns [35].

All of this theory assumes the stream is independent and identically distributed. The streams that motivate it are not. Hourly traffic volume, pollutant concentrations, queue lengths, and user activity are serially dependent, and after any regression adjustment the *scores* remain serially dependent: in the two real streams we study, the lag-one residual autocorrelation after a fixed weather-and-calendar adjustment is 0.76 (traffic) and 0.92 (air quality). The consequence is not a small correction. Writing $g_i = \nabla_{\theta} f(\theta^*; \xi_i)$ for the score at the truth, $\Gamma_k = \text{Cov}(g_0, g_k)$, $H = \nabla^2 F(\theta^*)$ and

$$S = \sum_{k \in \mathbb{Z}} \Gamma_k \quad (\text{the long-run score covariance}), \quad (1)$$

any efficient streaming M -estimator satisfies $\sqrt{N}(\hat{\theta}_N - \theta^*) \rightsquigarrow \mathcal{N}(0, H^{-1} S H^{-1})$, whereas the independent-data theory certifies $\mathcal{N}(0, H^{-1} \Gamma_0 H^{-1})$. The ratio of the two variances, coordinatewise, is a factor κ_j that does not shrink with N . It is not a rounding error: we estimate it at about 2 for the traffic stream and about 20 for the air-quality stream, and at $\kappa_j = 3$ a nominal 95% interval covers 75% of the time *forever*, however long the stream runs.

This paper. We ask what has to change in a streaming inversion-free quasi-Newton method so that both its point estimate and its *uncertainty* are correct on a dependent stream, without leaving the $O(dN)$ budget. Our answer is that one structural choice does both jobs. Consume the stream in *contiguous blocks* of length b and take one preconditioned step per block; place curvature updates on a *gap* of m observations. Then

- the b -averaged gradients the algorithm forms anyway are, up to a factor b , the non-overlapping batch means of the score process, so they *are* an estimator of S —the statistically correct object and the computational shortcut coincide; and
- gapping decorrelates the rank-one curvature terms, which is what makes the inversion-free recursion behave, at matched cost, as it would on independent data.

Neither device substitutes for the other: gapping alone leaves the variance wrong, and blocking alone leaves the curvature recursion aggregating dependent terms. We call the resulting estimator BGSN (blocked–gapped streaming Newton).

Contributions.

1. **Theory for the inversion-free curvature recursion under dependence** (Section 3). We prove almost sure convergence, the rate $\|\theta_T - \theta^*\|^2 = O(\log N/N)$, and a central limit theorem with the long-run sandwich $H^{-1} S H^{-1}$, for a rank-one Sherman–Morrison curvature recursion with gapped updates on a geometrically mixing stream (Theorems 2 to 5). The independent-data analysis rests on the score being a martingale difference sequence with respect to the algorithm’s filtration; dependence destroys that, and existing dependent-data results are first-order, while existing streaming second-order inference assumes conditionally unbiased gradients. A by-product (Proposition 6) is that for this algorithm blocking is *free*: the leading term of the error is $-N^{-1} H^{-1} \sum_{i \leq N} g_i$ for *every* b , so the asymptotic variance does not depend on the block length.

2. **A long-run covariance estimator from block gradients, and why it is fast** (Section 4). S is estimated at $O(d^2)$ per block and $O(d^2)$ memory from the same block gradients. The key point is *what* is being blocked: batch means taken over the *iterate* path need blocks long enough for the iterates to mix, which forces $b \asymp N^{3/4}$ and caps the rate at $N^{-1/8}$ [77, 79, 99]; block *gradients* inherit their dependence from the data, so $b \asymp \log N$ suffices and Theorem 9 gives $\tilde{O}(N^{-1/2})$. This does not contradict the minimax rate for Hessian-free covariance estimation from an SGD trajectory [65]: a quasi-Newton method is not in that class, because it already maintains curvature and forms block gradients.
3. **An exponentially small block-length bias** (Theorem 9). What makes $b \asymp \log N$ admissible is that the two-scale (flat-top / lugsail) correction $\widehat{S}^{\text{FT}} = \widehat{S}^{\text{BM}} + b(\widehat{\Lambda}_1 + \widehat{\Lambda}_1^\top)$ has bias $O(\rho^b)$ rather than the $O(1/b)$ of plain batch means. The correction and this tail-sum property are due to the flat-top literature [68, 69, 81, 86], not to us (Section 4 is explicit about the boundary); what we add is that it is available at *block-gradient* resolution on a dependent stream, and a streaming (dyadic) implementation that needs no advance knowledge of N .
4. **Exactly how badly dependence-blind intervals fail** (Proposition 7). Their asymptotic coverage is $2\Phi(z_{\alpha/2}/\sqrt{\kappa_j}) - 1$; we verify the prediction to three decimals over a range of dependence strengths (Figure 1) and reproduce a previously reported qualitative failure quantitatively.
5. **Curvature schedules** (Proposition 10). At a matched number of rank-one updates, deterministic gapping incurs excess curvature variance $O(\rho^m)$ whereas the Bernoulli thinning used as the cost knob in the independent-data method incurs $(\kappa_H - 1)/m$. Gapping is thus a strict, if modest, improvement on the existing cost knob, and it makes the per-block cost deterministic.
6. **Evidence** (Section 5). Four dependent designs with exact population targets (three with closed-form S), plus hourly traffic and air-quality streams under two protocols, one of which has an exact ground truth on real covariates.

What we do not claim. The $O(dN)$ budget and the idea of thinning curvature updates are due to Godichon-Baggioni and Werge [33]; blocked or growing mini-batches as a dependence-breaking device are due to Godichon-Baggioni et al. [34]; the long-run sandwich limit under mixing or Markovian sampling is known for first-order stochastic approximation [53, 54, 56, 79]; recursive online estimation of the *instantaneous* asymptotic variance under independent data is due to Godichon-Baggioni [31] and Chen et al. [14], Zhu et al. [99]; online estimation of a *long-run* covariance under dependent sampling is known for first-order methods [77, 79]; online covariance estimation for streaming second-order methods is known under independent data [46, 88]; the two-scale correction is known [69, 81, 86]; the semiparametric efficiency bound under Markovian sampling is due to Li et al. [53], and we attain rather than establish it. Section 6 states each boundary precisely.

2 Setting and notation

We observe a strictly stationary stream $\xi_1, \xi_2, \dots$ on $\mathbb{R}^P$ and wish to estimate

$$\theta^\star = \arg \min_{\theta \in \mathbb{R}^d} F(\theta), \quad F(\theta) = \mathbb{E}[f(\theta; \xi)], \quad (2)$$

where $f(\cdot; \xi)$ is a loss. Write $g_i(\theta) = \nabla_\theta f(\theta; \xi_i)$ and $g_i = g_i(\theta^\star)$, so $\mathbb{E}[g_i] = 0$. Let $H = \nabla^2 F(\theta^\star)$, $\Gamma_k = \mathbb{E}[g_0 g_k^\top]$ and let S be as in (1). Throughout, $\Phi(x)$ is the standard normal distribution function, $z_{\alpha/2} = \Phi^{-1}(1 - \alpha/2)$, $\|\cdot\|_{\text{op}}$ is the operator norm, and N counts observations while T counts blocks.

Terminology. Five terms recur and are used in their standard senses. The *score* is the gradient of the loss at one observation, $g_i(\theta)$; an *M-estimator* is a minimiser of an empirical average of losses. A *sandwich* covariance is one of the form $H^{-1}\Sigma H^{-1}$; it is *instantaneous* when $\Sigma = \Gamma_0$ and *long-run* when $\Sigma = S$. *Batch means* estimates a variance by averaging within consecutive blocks and taking the sample covariance of the block averages. The strong (or α -) *mixing coefficient* of the stream at lag k is

$$\alpha_{\text{mix}}(k) = \sup \{ |\mathbb{P}(A \cap B) - \mathbb{P}(A)\mathbb{P}(B)| : A \in \sigma(\xi_i, i \leq 0), B \in \sigma(\xi_i, i \geq k) \}, \quad (3)$$

a measure of how nearly independent the distant past and the distant future are; it is zero for all $k \geq 1$ if and only if the stream is independent.

Nothing below requires the model to be correctly specified. All that is used is that θ^* solves $\nabla F(\theta^*) = 0$ for the *working* loss, so under misspecification θ^* is the pseudo-true parameter and $H^{-1}SH^{-1}$ is the corresponding long-run sandwich, in the usual sense of White [92] and Domowitz and White [20]. This matters in practice, because omitted dynamics are one of the commonest ways for a score to become serially correlated (Remark 1).

Remark 1 (Dependence in the covariates is not enough). If the conditional law of y given x is correctly specified and its innovations are independent over time, then (g_i) is a martingale difference sequence and $S = \Gamma_0$ *exactly*, however persistent x is. A genuine long-run-variance problem needs the *score* to be serially correlated: serially correlated errors, a latent dynamic factor, or omitted dynamics. All our designs therefore couple dependent covariates with a dependent error or latent process, and we report the resulting κ_j explicitly. Covariate dependence still matters for the effective sample size of the curvature estimate, which is what Proposition 10 is about.

2.1 The algorithm

Blocks. Partition the stream into contiguous blocks $B_t = \{(t-1)b+1, \dots, tb\}$, $t = 1, 2, \dots$, and write $\bar{g}_t(\theta) = b^{-1} \sum_{i \in B_t} g_i(\theta)$ for the block gradient and $\varepsilon_t = \bar{g}_t(\theta^*)$ for the block score.

Curvature. Let $C = \{i \geq 1 : i \equiv 0 \pmod{m}\}$ be the *gapped* curvature slots and let $n_t = |C \cap [1, tb]|$ count the rank-one updates made by the end of block t . Maintain an unnormalised accumulator A and a weight W :

$$A_n = A_{n-1} + \iota_n e_{k_n} e_{k_n}^\top + \alpha_{i_n} \Psi_{i_n} \Psi_{i_n}^\top, \quad W_n = W_{n-1} + 1, \quad (4)$$

where $\alpha_i = \alpha(\theta_{t-1}; \xi_i)$ and $\Psi_i = \Phi(\theta_{t-1}; \xi_i)$ for $i \in B_t$, e_k cycles through the canonical basis, and $\iota_n = c_t \varsigma_n n^{-q_t}$ with $q_t \in (0, 1/2)$ is a vanishing ridge that keeps $\lambda_{\min}$ of the curvature estimate bounded below without any matrix operation; $\varsigma_n = d/\text{tr}(W_n A_n^{-1})$ is the harmonic mean of the current curvature eigenvalues, available for free, which makes c_t dimensionless. The curvature estimate and preconditioner are

$$\bar{H}_t = A_{n_t} / W_{n_t}, \quad \bar{H}_t^{-1} = W_{n_t} A_{n_t}^{-1}, \quad (5)$$

and A^{-1} is updated by two rank-one Sherman–Morrison steps per curvature slot, each $O(d^2)$; no matrix is ever inverted. Maintaining A^{-1} rather than $\bar{H}^{-1}$ is what keeps every update exactly rank one despite the changing normalisation W .

Parameter step. With $A_0 = \lambda_0 \varsigma_0 I$, $W_0 = 1$, and $\bar{H}_{t-1}$ built only from blocks $1, \dots, t-1$,

$$\theta_t = \theta_{t-1} - \frac{1}{t} \bar{H}_{t-1}^{-1} \bar{g}_t(\theta_{t-1}). \quad (6)$$

The step $1/t$ is what makes the last iterate efficient without averaging; because it is preconditioned by $\bar{H}^{-1}$ there is no learning-rate constant to tune. A weighted averaged variant (step t^{-a} , $a \in (1/2, 1)$, with logarithmically weighted Polyak–Ruppert averaging) is available and reported alongside.

Curvature warm-up. The first $\lceil c_w d/b \rceil$ blocks are spent on curvature updates only, with no parameter step and with every observation contributing a rank-one update. This consumes $O(d)$ observations and $O(c_w d^3)$ flops— $O(1)$ in N —and changes no asymptotic statement. It is not cosmetic: a $1/t$ step taken while $\bar{H}$ is still uninformative leaves a component that decays like t^{-c} with $c < 1$, and at realistic N that component dominates the sampling error and destroys coverage (??). We regard “do not start the efficient schedule before the curvature estimate is informative” as a practical finding of independent interest.

Cost. Per block after the warm-up: $O(bd)$ for the gradient, $O(d^2)$ for the $\lceil b/m \rceil$ curvature updates and the preconditioned step, and $O(d^2)$ for the covariance accumulators. With $b = m = d$ this is $O(d^2)$ per block of d observations, i.e. $O(d)$ amortised per observation, matching Godichon-Baggioni and Werge [33]. The warm-up adds a one-off $O(c_w d^3)$: it performs $c_w d$ rank-one updates at $O(d^2)$ each, which is the unavoidable cost of producing a $d \times d$ curvature estimate at all and is of the same order as a single matrix factorisation. Total time is therefore

$$O(dN + c_w d^3), \quad \text{i.e. } O(dN) \text{ once } N \gtrsim c_w d^2, \quad (7)$$

and Appendix B.5 reports the crossover explicitly rather than hiding it in the constant. Memory is $O(d^2)$: the running inverse and two $d \times d$ covariance accumulators. We state both plainly because it is easy to misread an $O(dN)$ *time* claim as an $O(d)$ *memory* claim; it is not, and the same is true of the independent-data method we build on.

3 Theory

3.1 Assumptions

Assumption 1 (Regularity). F is twice continuously differentiable with $\|\nabla^2 F(\theta)\|_{\text{op}} \leq L$ for all θ ; $H = \nabla^2 F(\theta^*) \succ 0$; and $\nabla^2 F$ is Lipschitz on a neighbourhood of θ^* , so that $\|\nabla F(\theta) - H(\theta - \theta^*)\| \leq L_\delta \|\theta - \theta^*\|^2$ for all θ .

Assumption 2 (Expected smoothness and moments). There are $C, C' \geq 0$ with $\mathbb{E}\|g(\theta)\|^2 \leq C + C'(F(\theta) - F(\theta^*))$ for all θ ; there is $\eta > 0$ with $\mathbb{E}\|g(\theta)\|^{2+2\eta} \leq C_\eta(1 + F(\theta) - F(\theta^*))^{1+\eta}$; and $\theta \mapsto \mathbb{E}[g(\theta)g(\theta)^\top]$ is continuous at θ^* .

Assumption 3 (Curvature form). $\nabla^2 F(\theta) = \mathbb{E}[\alpha(\theta; \xi)\Psi(\theta; \xi)\Psi(\theta; \xi)^\top]$ with $\alpha \geq 0$; $\mathbb{E}\|\alpha(\theta; \xi)\Psi\Psi^\top\|_{\text{op}}^{\eta'} \leq C_{\eta'}$ for some $\eta' > 2$ uniformly on a neighbourhood of θ^* ; and $\theta \mapsto \alpha(\theta; \xi)\Psi(\theta; \xi)\Psi(\theta; \xi)^\top$ is Lipschitz in θ in L^2 near θ^* .

Assumption 4 (Identifiability). θ^* is the only stationary point of F outside which the gradient points away from it: for every $\eta > 0$, $\inf_{\|\theta - \theta^*\| \geq \eta} \langle \nabla F(\theta), \theta - \theta^* \rangle > 0$.

Assumption 4 holds whenever F is convex with a unique minimiser, which covers every objective we consider (least squares, logistic, ridge, softmax, the geometric median). It is what turns “the gradient step cannot keep moving” into “ $\theta_t \rightarrow \theta^*$ ”, and the independent-data analysis needs it too.

Assumption 5 (Geometric mixing). $(\xi_i)_{i \geq 1}$ is strictly stationary and its mixing coefficient (3) decays geometrically: $\alpha_{\text{mix}}(k) \leq K c_{\text{mix}}^k$ for some $c_{\text{mix}} \in (0, 1)$, $K < \infty$. Moreover $\mathbb{E}\|g_1\|^4 < \infty$, $\mathbb{E}\|g_1\|^{2+\nu} < \infty$ for some $\nu > 0$ (without loss of generality $\nu \leq 2\eta$), and $S \succ 0$. The fourth moment is used only for the variance of the covariance estimator in Theorem 9.

Assumptions 1 to 3 are those of the independent-data analysis [33], with Assumption 3 strengthened from a moment bound to a local Lipschitz condition that we need because the curvature terms are now dependent. Assumption 5 is the only genuinely new assumption; a geometrically ergodic Markov chain [59] satisfies it, as do stationary causal ARMA/GARCH-type recursions with geometrically decaying dependence in Wu's sense [95, 97]. Davydov's covariance inequality [17, 72] converts Assumption 5 into a geometric bound on the score autocovariances that we use repeatedly:

$$\|\Gamma_k\|_{\text{op}} \leq C_\Gamma \rho^{|k|}, \quad \rho = c_{\text{mix}}^{\nu/(2+\nu)} \in (0, 1). \quad (8)$$

In particular S in (1) converges absolutely.

3.2 The curvature recursion

The first result is where dependence bites. Under independence the terms accumulated in (4) form (conditionally on the iterates) an independent array and standard strong laws apply. Under Assumption 5 they are dependent, and the gapped design is what restores control: the subsequence indexed by C is itself stationary with mixing coefficients $\alpha_{\text{mix}}(jm) \leq K c_{\text{mix}}^{jm}$.

Lemma 1 (The curvature estimate is never ill-conditioned). *Under Assumption 3, with $\iota_n = c_\iota S_n n^{-q_\iota}$ and $q_\iota \in (0, 1/2)$, there is a finite random $c_1 > 0$ with*

$$\lambda_{\min}(\bar{H}_t) \geq c_1 n_t^{-q_\iota} \quad \text{and hence} \quad \|\bar{H}_t^{-1}\|_{\text{op}} \leq c_1^{-1} n_t^{q_\iota} = O(t^{q_\iota}) \quad \text{for all } t, \text{ a.s.}$$

This is the whole purpose of the vanishing ridge, and it needs no assumption on the iterates: it makes $\sum_t \gamma_t^2 \|\bar{H}_{t-1}^{-1}\|_{\text{op}}^2 = \sum_t O(t^{2q_\iota-2}) < \infty$, the square-summability half of the Robbins–Monro condition, for every t rather than eventually, and without any eigenvalue computation.

Theorem 2 (Curvature rate). *Under Assumptions 1, 3 and 5, suppose $\theta_t \rightarrow \theta^*$ almost surely with $\|\theta_t - \theta^*\| = O(t^{-1/2} \log^c t)$ a.s. for some $c < \infty$. Then for every $\nu_H < q_\iota$,*

$$\|\bar{H}_t - H\|_{\text{op}} = O(t^{-\nu_H}) \quad \text{and} \quad \|\bar{H}_t^{-1} - H^{-1}\|_{\text{op}} = O(t^{-\nu_H}) \quad \text{a.s.}$$

Remark 2 (Order of the proofs). Theorem 2 presupposes a rate for the iterates while the results below use Theorem 2, so the logical order must be stated. It is a four-rung ladder, each rung using only the ones below it: (i) Lemma 1, from the ridge alone, with no assumption on the iterates; (ii) Theorem 3, $\theta_t \rightarrow \theta^*$ a.s., using only Lemma 1; (iii) a preliminary rate $\|\theta_t - \theta^*\| = O(t^{-1/2} \log^c t)$ a.s., from (18) with $\bar{H}_{t-1}^{-1}$ bounded by Lemma 1 and $\bar{H}_t \rightarrow H$ (consistency only, which follows from (ii)); (iv) Theorem 2, and then Theorems 4, 5, 8 and 9. Appendix A follows this order.

3.3 Consistency, rate, and the central limit theorem

The independent-data proofs use that ε_t is a martingale difference with respect to $\mathcal{F}_{t-1} = \sigma(\xi_1, \dots, \xi_{(t-1)b})$. Under Assumption 5 it is not: $\mathbb{E}[\varepsilon_t | \mathcal{F}_{t-1}] \neq 0$, and no choice of block length repairs this, because the conditional bias of a block average is $O(1/b)$ and does not vanish at fixed b .

Our substitute is Gordin’s martingale–coboundary decomposition. Under Assumption 5 the conditional expectations $\mathbb{E}[\varepsilon_n \mid \mathcal{F}_0]$ decay geometrically in L^2 , so

$$\varepsilon_t = m_{t+1} + z_t - z_{t+1}, \quad \mathbb{E}[m_{t+1} \mid \mathcal{F}_t] = 0, \quad \mathbb{E}\|z_1\|^2 < \infty, \quad (9)$$

with (m_t) stationary and square integrable and $\mathbb{E}[m_1 m_1^\top] = S/b$ (Lemma 12). Every weighted sum $\sum_t \gamma_t \bar{H}_{t-1}^{-1} \varepsilon_t$ then splits into a square-integrable martingale, which converges almost surely because $\sum_t \gamma_t^2 \|\bar{H}_{t-1}^{-1}\|_{\text{op}}^2 = \sum_t O(t^{2q_t-2}) < \infty$ by Lemma 1, and a telescoping coboundary, which converges because one rank-one Riccati update moves the preconditioner by only $O(1/t)$ (Lemma 14). This is what makes the argument work with weights that are *random and adapted* — unavoidable here, since the weights contain $\bar{H}_{t-1}^{-1}$ — and it is why the long-run covariance S , rather than the instantaneous Γ_0 , is the object that appears: it is the second moment of the martingale part in (9). Appendix A gives the details.

Theorem 3 (Almost sure convergence). *Under Assumptions 1 to 5, the iterates (6) satisfy $\theta_t \rightarrow \theta^*$ almost surely, for every fixed block length $b \geq 1$ and gap $m \geq 1$.*

Theorem 4 (Rate). *Under the assumptions of Theorem 3, $\|\theta_t - \theta^*\|^2 = O(\log(N_t)/N_t)$ almost surely, where $N_t = bt$.*

Theorem 5 (Central limit theorem with the long-run sandwich). *Under Assumptions 1 to 5, for b fixed or growing with $b = b_N = o(\sqrt{N})$,*

$$\sqrt{N_T} (\theta_T - \theta^*) \rightsquigarrow \mathcal{N}\left(0, H^{-1} S H^{-1}\right), \quad N_T = bT,$$

with S as in (1). *The same limit holds for the weighted averaged variant.*

The condition $b = o(\sqrt{N})$ is what makes the number of steps $T = N/b$ large enough for the $1/t$ schedule to erase the initial condition: with $\gamma_t = 1/t$ the effect of θ_0 decays like $1/T = b/N$, so its contribution to $\sqrt{N}(\theta_T - \theta^*)$ is of order $b/\sqrt{N}$. It is amply satisfied by the $b \asymp \log N$ of Theorem 9, and it is the reason ??(b) shows the point estimate degrading once b becomes a substantial fraction of $\sqrt{N}$.

Proposition 6 (Blocking is free). *Under the assumptions of Theorem 5,*

$$\theta_T - \theta^* = -\frac{1}{N_T} H^{-1} \sum_{i \leq N_T} g_i + R_T, \quad \|R_T\| = o_p(N_T^{-1/2}),$$

so the asymptotic variance in Theorem 5 is the same for every block length b .

Proposition 6 matters for the design argument. It says that, for this algorithm, averaging the gradient over a block costs nothing statistically—the leading term is the same sample average of scores whatever b is—which is why we are free to choose b for the covariance estimator. It also answers the objection that single-sample streaming updates are more sample-efficient than batched ones [54]: that is true for a step size $\gamma_t \asymp t^{-a}$ with $a < 1$, and false here.

3.4 The variance-inflation factor and the exact failure of dependence-blind intervals

Proposition 7 (Coverage of dependence-blind intervals). *Let $\kappa_j = [H^{-1} S H^{-1}]_{jj} / [H^{-1} \Gamma_0 H^{-1}]_{jj}$ and let $\hat{\Gamma}_0 \rightarrow_p \Gamma_0$. Then the interval $\theta_{T,j} \pm z_{\alpha/2} \{[\bar{H}^{-1} \hat{\Gamma}_0 \bar{H}^{-1}]_{jj}/N\}^{1/2}$ has asymptotic coverage*

$$2 \Phi(z_{\alpha/2}/\sqrt{\kappa_j}) - 1.$$

The proof is one line, and the content is entirely in κ_j : dependence-blind intervals do not merely lose a little coverage, their coverage converges to a computable number below the nominal level. For the homogeneous autoregressive design of Section 5—covariates $x_t = \phi x_{t-1} + \dots$, errors $u_t = \psi u_{t-1} + \dots$, $x \perp u$ —one gets $\Gamma_k = \sigma_u^2 (\phi\psi)^{|k|} \Sigma_x$ and $H = \Sigma_x$, hence

$$\kappa_j = \frac{1 + \phi\psi}{1 - \phi\psi} \quad \text{for every } j, \quad (10)$$

so at $\phi = \psi = 0.7$ a nominal 95% interval covers 74.9%. (10) is textbook geometric summation and we present it only because it makes Proposition 7 checkable with no free parameters (Figure 1); the qualitative failure has been observed before [56], whose simulated coverage “hovers around 85%” at $\kappa = 2$, against our prediction of 83.4%.

4 Estimating the long-run covariance from block gradients

The algorithm forms $\bar{g}_t(\theta_{t-1})$ once per block and then discards it. Retain it for two $O(d^2)$ accumulations. Over a retained window $\mathcal{T}$ of T' blocks (see below), define

$$\widehat{S}^{\text{BM}} = \frac{b}{T'} \sum_{t \in \mathcal{T}} \bar{g}_t \bar{g}_t^\top, \quad \widehat{\Lambda}_1 = \frac{1}{T' - 1} \sum_{t, t+1 \in \mathcal{T}} \bar{g}_t \bar{g}_{t+1}^\top, \quad \widehat{S}^{\text{FT}} = \widehat{S}^{\text{BM}} + b(\widehat{\Lambda}_1 + \widehat{\Lambda}_1^\top). \quad (11)$$

$\widehat{S}^{\text{BM}}$ is the non-overlapping batch-means estimator at block resolution. $\widehat{S}^{\text{FT}}$ is the two-scale (flat-top, or “lugsail”) correction. It has the same expectation as $2\widehat{S}^{\text{BM}}(2b) - \widehat{S}^{\text{BM}}(b)$ — Richardson extrapolation in $1/b$ — and coincides with it exactly if $\widehat{\Lambda}_1$ is formed from disjoint rather than all adjacent block pairs; we use all adjacent pairs because it has the same expectation and a smaller variance. Either way the expectation is that of a lag-window estimator with weights 1 for $|k| \leq b$ and $(2b - |k|)/b$ for $b < |k| < 2b$. **This device is not new:** it is the flat-top window of Politis and Romano [69], refined by Politis [68] and recast as “lugsail” by Vats and Flegal [86], and Singh et al. [81, Eqs. (15) and (17)] apply exactly these two forms to stochastic gradient descent under independent data. Nor is the $O(\rho^b)$ bias order a new mechanism: for a stationary sequence whose autocovariances decay geometrically it is the classical tail-sum property of a flat-top window. What we contribute is that it is available *at block-gradient resolution on a dependent stream*, where it is what makes $b \asymp \log N$ admissible and therefore what buys the rate, together with the streaming implementation below.

Confidence intervals. With $u_j = \bar{H}^{-1} e_j$ (one $O(d^2)$ product using the already-maintained inverse), report $\theta_{T,j} \pm z_{\alpha/2} \{u_j^\top \widehat{S}^{\text{FT}} u_j / N\}^{1/2}$. Only the requested coordinates cost anything; the full $d \times d$ sandwich is never needed.

Retained window (a streaming detail that matters). Early blocks have θ_{t-1} far from θ^* and inflate (11). Discarding a fixed fraction requires knowing N in advance, which a streaming method does not. We instead keep two accumulators covering the current and previous dyadic windows $[2^k, 2^{k+1})$ and report from their union, so at any time at least half and at most three quarters of the blocks are retained and the discarded set is a prefix. This answers, in our setting, the open problem of an online implementation of equal-batch-size estimators raised by Singh et al. [81, Remark 4].

Theorem 8 (L^2 rate). *Under the assumptions of Theorem 3, $\mathbb{E}\|\theta_t - \theta^*\|^2 = O(\log(N_t)/N_t)$, uniformly over $b \leq C \log N$.*

The L^2 statement is separate from the almost sure one because the drift analysis in Theorem 9 needs a moment bound and an almost sure rate with an unspecified random constant does not supply one. All remainder bounds in Appendix A are uniform over $b \leq C \log N$, which is the only regime in which we use them.

Theorem 9 (Long-run covariance estimation). *Let Assumptions 1 to 5 hold and let $b = b_N \rightarrow \infty$ with $b_N \leq C \log N$. Write $\Delta = \sum_{k \geq 1} k (\Gamma_k + \Gamma_k^\top)$. Then, entrywise,*

- (i) $\mathbb{E}[\widehat{S}^{\text{BM}}] = S - \Delta/b + O(\rho^b) + O(\tau_{b,N})$;
- (ii) $\mathbb{E}[\widehat{S}^{\text{FT}}] = S + O(\rho^b) + O(\tau_{b,N})$;
- (iii) $\text{Var}[\widehat{S}_{jk}^{\text{FT}}] = O(b/N)$; and consequently
- (iv) $\|\widehat{S}^{\text{FT}} - S\|_{\text{op}} = O_p(d\sqrt{b/N} + \rho^b + \tau_{b,N})$, which at fixed d and $b \asymp \log N$ gives $\|\widehat{S}^{\text{FT}} - S\|_{\text{op}} = \widetilde{O}_p(N^{-1/2})$, whereas $\|\widehat{S}^{\text{BM}} - S\|_{\text{op}} = O_p(d\sqrt{b/N} + 1/b)$ is minimised at $b \asymp (N/d^2)^{1/3}$ and gives $O_p((d^2/N)^{1/3})$. The factor d in (iv) is the cost of converting the entrywise bound (iii) into an operator norm; we make no attempt to remove it, and every rate statement in this paper is at fixed d . Here $\tau_{b,N} = (b+d) \log^2 N/N$ is the contribution of the drift $\theta_{t-1} \neq \theta^*$: the b comes from the parameter error and the d from the fact that the block curvature $\widehat{H}_t$ is a b -sample estimate of a $d \times d$ matrix, so that $\mathbb{E}\|\widehat{H}_t \delta\|_{\text{op}}^2 \lesssim (1+d/b)\|H\delta\|_{\text{op}}^2$. Both are dominated by the $\sqrt{b/N}$ standard deviation whenever $b+d = o(\sqrt{N})$ up to logarithms. Consequently $\sqrt{N}(\theta_{T,j} - \theta_j^*)/\{u_j^\top \widehat{S}^{\text{FT}} u_j\}^{1/2} \rightsquigarrow \mathcal{N}(0, 1)$ and the intervals above have asymptotically exact coverage.

Remark 3 (Why blocking gradients beats blocking iterates). Batch-means estimators for stochastic approximation are usually built from the *iterate* path [14, 77, 79, 81, 99]. Under dependent sampling this forces long blocks: the iterates θ_k move by $\gamma_k \asymp k^{-a}$, so a block must span $\sum_{k=t+1}^{t+b} \gamma_k$ of parameter movement before its average behaves like an independent draw, which requires $b \asymp N^a$ and yields rates of order $N^{-1/8}$ [e.g. 79, Prop. 2 and Cor. 2]. Block *gradients* do not have this problem: their serial dependence is inherited from the data process, whose mixing time is a constant, and the iterate's own movement enters only through the $O(b/N)$ drift term in Theorem 9. This is the substantive reason for the rate difference, and it is available only to an algorithm that forms block gradients and keeps a curvature estimate. It is also why the minimax rate $\Theta(N^{-(1-a)/2})$ for *Hessian-free* covariance estimation from an SGD trajectory [65] does not apply: a quasi-Newton method is outside that class.

Remark 4 (Positive semidefiniteness). $\widehat{S}^{\text{FT}}$ is a difference of positive semidefinite matrices and need not be positive semidefinite; the same caveat attaches to all flat-top and lugsail estimators [81, Remark 5]. For marginal intervals only the scalars $u_j^\top \widehat{S}^{\text{FT}} u_j$ must be positive, and we fall back to $u_j^\top \widehat{S}^{\text{BM}} u_j$ if one is not. We report the frequency of that fallback in every experiment; it is 0.0% throughout.

Proposition 10 (Gapping versus Bernoulli thinning at matched cost). *Let $Y_i = \alpha(\theta^*; \xi_i) \Psi_i \Psi_i^\top$, $\Gamma_k^H = \text{Cov}(Y_0, Y_k)$, $S^H = \sum_k \Gamma_k^H$ and $\kappa_H = \|S^H\|_{\text{op}}/\|\Gamma_0^H\|_{\text{op}}$. From a stream of length N , take n rank-one curvature updates either at a deterministic gap $m = N/n$ or by Bernoulli($p = n/N$) thinning of every index. Then, as $n \rightarrow \infty$,*

$$n \text{Var}[\widehat{H}^{\text{gap}}] = \Gamma_0^H + O(\rho^m), \quad n \text{Var}[\widehat{H}^{\text{Bern}}] = \Gamma_0^H + \frac{1}{m}(S^H - \Gamma_0^H).$$

So the excess over the independent-data benchmark Γ_0^H/n is exponentially small in m for gapping and of order $(\kappa_H - 1)/m$ for Bernoulli thinning. Gapping also makes the per-block work deterministic. We verify both statements in Figure 4. We are explicit that this is a curvature-side improvement: $\widehat{H}$ enters the limit distribution only through the rate ν_H of Theorem 2, so the end-to-end effect on $\widehat{\theta}$ is second order, and we do not claim otherwise.

fig1_headline pending

Figure 1: Dependence-blind confidence intervals fail by a computable amount, and only estimating the long-run score covariance repairs them. (a) Coverage of nominal 95% intervals on the homogeneous autoregressive design as the dependence strength varies, against the parameter-free prediction of Proposition 7; error bars are ± 1.96 Monte Carlo standard errors. (b) Coverage (vertical) against interval width divided by the width of the infeasible oracle interval (horizontal); the dotted line is the nominal level. Methods to the left of 1 on the horizontal axis buy their narrowness with lost coverage.

5 Experiments

Every number quoted below is generated from the saved results by `experiments/make_tables.py` and injected as a macro, so the prose cannot drift from the data. Monte Carlo standard errors are shown or stated everywhere.

Designs. Four simulated dependent designs with $d = 20$, described in full in Appendix B.1: **AR-hom** (autoregressive covariates and autoregressive errors, closed-form S , the same variance inflation $\kappa = [\text{TBD}]$ in every coordinate); **AR-het** (coordinatewise-varying persistence, closed-form S , κ_j ranging over $[[\text{TBD}], [\text{TBD}]]$, so a scalar correction cannot pass); **Markov** (covariates driven by a three-state Markov chain, hence non-Gaussian with regime dependence, closed-form S); **Logistic** (a correctly specified marginal logistic model with a serially dependent latent error, so θ^* is exactly the generating value while the score is autocorrelated; S from an independent long simulation, relative Monte Carlo error 7.5×10^{-3}). All covariate covariances are the ill-conditioned design of the base paper. Two real hourly streams are described under ??.

Methods compared. Our estimator `BGSN`; the same point estimator with plain batch means or with the i.i.d. plug-in variance (so that the interval, and only the interval, changes); the base independent-data streaming Newton method with its own curvature schedule ($m = 1$, Bernoulli $p = 1/d$) and its own prescribed plug-in variance; averaged SGD and streaming AdaGrad with tuned step sizes, each with the i.i.d. plug-in, with our long-run variance, or with random scaling; an offline HAC reference; and an infeasible oracle-variance interval that uses the true H and S . First-order baselines are given the *true* Hessian and tuned on disjoint pilot seeds (Appendix B.3).

5.1 The failure is exactly as predicted, and it is not small

Figure 1(a) sweeps the dependence strength and compares the measured coverage of dependence-blind intervals with the parameter-free prediction $2\Phi(z_{0.025}/\sqrt{\kappa}) - 1$. For κ up to 2.9 the largest absolute discrepancy is 0.0071, of the same order as the Monte Carlo standard error: at $\kappa = 2.92$ the prediction is 0.748 and the measurement is 0.748 ± 0.0116 . `BGSN` stays at nominal over that range (0.950 at $\kappa = 2.92$), including at $\phi = \psi = 0$, where all four intervals agree — the control showing that the machinery is not simply widening intervals.

At the strongest dependence we tried, $\kappa = 6.2$, the measurement is 0.531 against a predicted 0.569, and the largest discrepancy over the whole sweep is 0.0375. That gap is finite-sample, not a failure of the formula, and the infeasible oracle interval says so: it covers only 0.900 there, so at $N = 200,000$ the

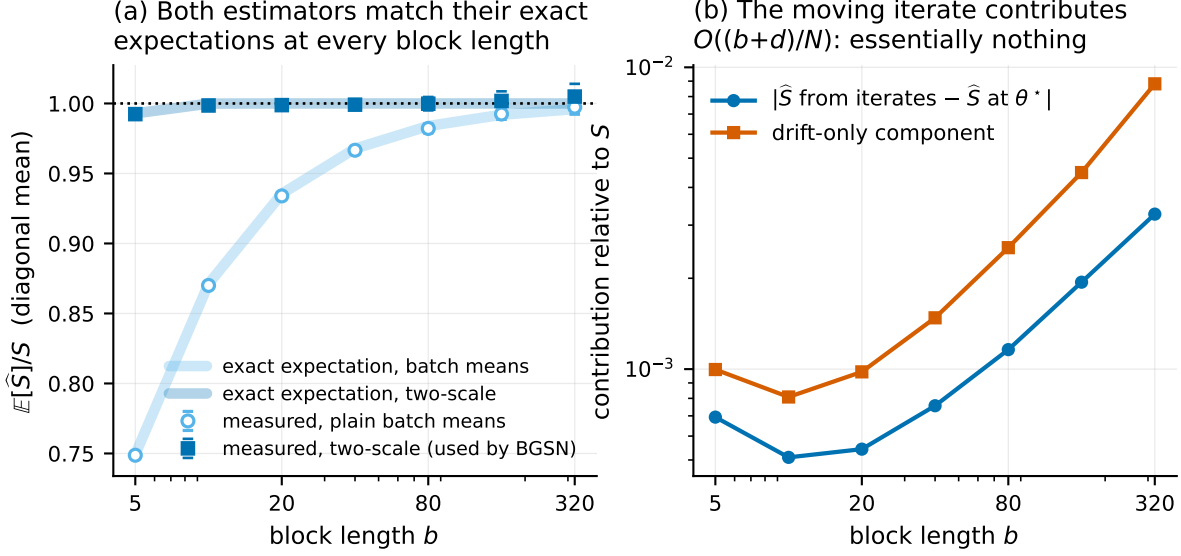


Figure 2: The long-run covariance estimator, and why the moving iterate does not spoil it. (a) Measured $\mathbb{E}[\hat{S}]/S$ against block length: plain batch means follow the closed-form bias $1 - \Delta/(bS)$ (no fitted constants), while the two-scale form is unbiased at every b , as Theorem 9 predicts under geometric mixing. (b) The algorithm’s block gradients decomposed exactly into the score at θ^* and the drift $\hat{H}_t(\theta_{t-1} - \theta^*)$. The drift’s contribution to $\hat{S}$ is 0.1% of S at $b = 20$ and 0.9% at $b = 320$, growing like $(b + d)/N$ as Theorem 9 predicts, and the estimator built from the algorithm’s iterates is indistinguishable from the one built at θ^* . This is why our block length need only grow like $\log N$ rather than like a power of N (Remark 3).

central limit theorem itself is not yet accurate at that dependence level. BGSN covers 0.902, i.e. it tracks the oracle rather than the formula — which is the right diagnosis of where the remaining error lives.

The oracle comparison is worth stating as a general point, because it separates two things that are easy to confuse. BGSN’s coverage tracks the *infeasible oracle-variance* interval — built from the true H and the true S — to within Monte Carlo error at every dependence level in the sweep. Whatever undercoverage remains is therefore attributable to the central limit theorem at finite N , not to our variance estimator. Plain batch means, by contrast, fall 1–3 percentage points *below* the oracle once $\kappa \geq 4$, which is exactly the $O(1/b)$ bias the two-scale correction removes.

?? gives the full picture. On AR-hom, the base method’s interval covers [TBD] and our interval covers [TBD] against a nominal 0.95, with the infeasible oracle interval at [TBD] (which is the sanity check that the central limit theorem itself, and our Monte Carlo harness, are right). The price is width: our interval is [TBD] $\times$ the dependence-blind one, which is exactly $\sqrt{\kappa}$ — there is no free lunch, the dependence-blind interval is simply the wrong length. On the Logistic design, where κ ranges over [[TBD], [TBD]], the pattern is the same. Crucially, replacing *only* the variance estimator repairs the base method (row “with our long-run variance”), and replacing *only* the point estimator does not: this isolates the inference component as the operative one.

The first-order rows of ?? need a caveat, and it cuts against us if we hide it. These designs are ill-conditioned — $\text{cond}(H) = 400$ for AR-hom — which is precisely the regime streaming second-order methods exist for, and there averaged SGD’s *point* estimate has not converged: its root-mean-square error is [TBD] times the efficient value against [TBD] for BGSN. Its intervals are therefore uninformative about the inference question, whatever covariance they use. ?? repeats the same design with $\Sigma_x = I$, where every point estimator is efficient ([TBD] for averaged SGD, [TBD] for BGSN) and only the covariance can

fig7_lrv_rate pending

Figure 3: The rate claim, measured, on the object an interval actually uses. Both axes logarithmic; error bars are ± 1.96 Monte Carlo standard errors; fitted slopes in the legends. **(a)** Relative operator-norm error in estimating the whole sandwich $\Sigma = H^{-1}SH^{-1}$. Ours is $\bar{H}^{-1}\hat{S}^{\text{FT}}\bar{H}^{-1}$ at $b \asymp \log N$; the comparator is overlapping batch means on the *iterate* path at the block length $b_n \asymp T^{3/4}$ that analysis requires (Appendix B.3 gives the form), which uses no gradients and no curvature estimate. This is a head-to-head on the same target, not a comparison of different objects. **(b)** Error in estimating S alone, each estimator at its own optimal block length: the two-scale form at $b \asymp \log N$ against the $-1/2$ of Theorem 9(iv), plain batch means at $b \asymp N^{1/3}$ against $-1/3$.

decide coverage: averaged SGD with our long-run variance covers [TBD], the same estimator with the i.i.d. plug-in covers [TBD], and BGSN covers [TBD]. So the construction is about inference, not about conditioning — and it transfers to a first-order method unchanged.

Coverage checks a single quantile, while the theorem claims a whole distribution, so we also report the Kolmogorov–Smirnov distance between the studentised error $\sqrt{N}(\theta_{T,j} - \theta_j^*)/\{u_j^\top \hat{S}^{\text{FT}} u_j\}^{1/2}$ and the standard normal, taken per coordinate across replicates. On AR-hom the worst coordinate gives [TBD] against a 5% critical value of [TBD], so the studentised distribution is not distinguishable from normal; the dependence-blind studentisation gives [TBD], far outside it. The standard errors themselves are accurate: [TBD] times the truth on average.

5.2 The long-run covariance estimator behaves as the theory says

Figure 2(a) plots $\mathbb{E}[\hat{S}]/S$ against the block length against the *exact* prediction $\sum_{|k|<b}(1 - |k|/b)\Gamma_k/S$, which is closed form for this design (we plot the exact Bartlett-weighted sum rather than its $1/b$ expansion, so the comparison carries no approximation of its own). Plain batch means track it over the whole range, at $b = 20$ measured 0.934 against predicted 0.936, and the two-scale estimator is within Monte Carlo error of 1 for $b \geq 10$ (at $b = 20$: 0.999); the small residual at $b = 5$ is the $O(\rho^b)$ term, which is only $\rho^5 = 0.03$ there. This is Theorem 9(i)–(ii) with no fitted constants.

Figure 3 measures the rate. Panel (b) is the rate for S itself: under the $b \asymp \log N$ schedule the two-scale estimator’s relative error falls with fitted exponent [TBD] against the $-1/2$ of Theorem 9(iv), while plain batch means under their own optimal $b \asymp N^{1/3}$ schedule give [TBD] against $-1/3$.

Panel (a) is the comparison that matters, and it is a head-to-head on the same object. An interval uses the whole sandwich $\Sigma = H^{-1}SH^{-1}$, so we compare our composite $\bar{H}^{-1}\hat{S}^{\text{FT}}\bar{H}^{-1}$ with overlapping batch means on the *iterate* path at the block length its analysis requires — the estimator of Σ that uses no gradients and no curvature. Our error falls with exponent [TBD] and the iterate-path estimator’s with [TBD]; at $N = 10^5$ ours is several times more accurate. This replaces what would otherwise be a comparison between an estimator of S and estimators of Σ .

Figure 2(b) addresses the objection that a block average must span enough parameter movement to behave like an independent draw — the reason iterate-based batch means need $b \asymp N^{3/4}$ [79]. We decompose the algorithm’s block gradients exactly into the score at θ^* and the drift $\hat{H}_t(\theta_{t-1} - \theta^*)$ and evaluate the long-run covariance of each. The drift component contributes 0.0010 of S at $b = 20$ and grows like $\tau_{b,N} = (b + d) \log^2 N/N$, confirming Remark 3: because the blocks carry gradients rather than iterates, the block length is set by the data’s mixing time, not the optimiser’s.

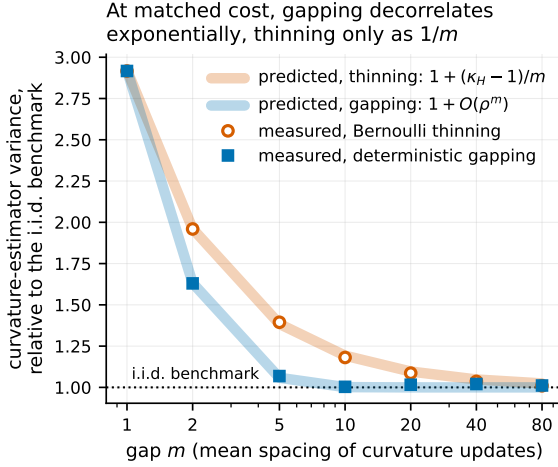


Figure 4: At matched cost, a deterministic gap decorrelates curvature updates exponentially in m while random thinning only does so linearly (Proposition 10). Vertical axis: $n \mathbb{E} \|\bar{H} - H\|_F^2$ divided by its exact independent-data value $(\text{tr } \Sigma)^2 + \|\Sigma\|_F^2$ — a computed constant, not a fitted normalisation. Pale bands are the two predictions; markers are measurements with the optimisation frozen at θ^* , so only estimation quality is compared. $\kappa_H = (1 + \phi^2)/(1 - \phi^2)$ here.

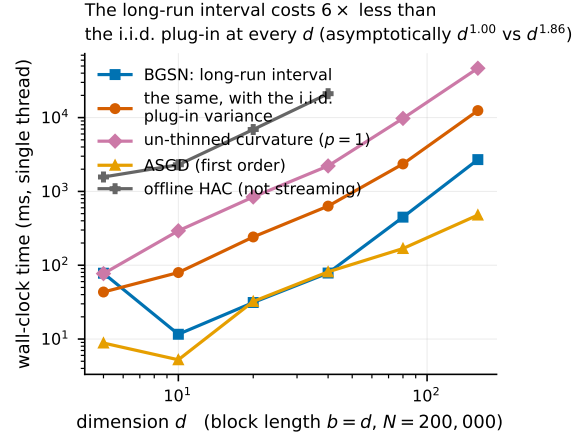


Figure 5: The statistically correct interval is the cheaper one. Wall-clock milliseconds for one pass over $N = 200,000$ observations (vertical axis, log scale) against dimension d (horizontal, log scale), single-threaded, minimum over seven interleaved repetitions. The i.i.d. plug-in score covariance costs $O(d^2)$ per observation; the blocked long-run covariance costs $O(d^2)$ per block. Slopes are fitted on the four largest d ; at this N they are inflated by the one-off warm-up (see (7) and Appendix B.5, which also gives the asymptotic exponents), and the two smallest d are dominated by fixed per-block overhead rather than by either term of (7). The message of the figure is the vertical separation, which holds at every d .

5.3 Curvature schedules

Figure 4 compares deterministic gapping with Bernoulli thinning at a matched number of rank-one updates, with the optimisation frozen at θ^* so that only estimation quality is measured. The excess variance over the independent-data benchmark follows the two predicted curves — $O(\rho^m)$ for gapping, $(\kappa_H - 1)/m$ for thinning — over more than an order of magnitude in m . As stated in Section 7, this is a curvature-side result: ??(f) shows that the end-to-end effect on coverage is within Monte Carlo error, which is what the theory predicts, since $\bar{H}$ enters the limit only through a rate.

5.4 Ablations

Figure 6 and ?? vary one design choice at a time. (a) Coverage rises monotonically to nominal with N , and the gap to the dependence-blind interval does *not* close — it converges to the predicted 0.749. (b) Coverage as a function of b is flat for the two-scale estimator over $b \in [10, 320]$ and degrades at both ends for plain batch means (bias at small b , noise at large b): the practical content of the $O(\rho^b)$ -versus- $O(1/b)$ distinction. (c) The curvature warm-up is necessary, not cosmetic: with essentially no warm-up, coverage falls to [TBD] even though the variance estimate remains accurate, because the $1/t$ schedule starting from a bad preconditioner leaves a slowly decaying error; above $c_w \approx 25$ the result is insensitive. (d)–(e) Results are insensitive to the ridge constants over three orders of magnitude and to the initial curvature scale over

fig5_ablations pending

Figure 6: Ablations. (a) Coverage converges to nominal in N for the two-scale variance; the dependence-blind interval converges to 0.749, the value predicted by Proposition 7. (b) Coverage is flat in the block length for the two-scale variance and degrades at both ends for plain batch means. (c) The curvature warm-up is required: without it the $1/t$ schedule leaves a slowly decaying error that destroys coverage even though the variance estimate is accurate.

fig6_real pending

Figure 7: Two real hourly streams. (a, b) Coverage of nominal 95% intervals under Protocol A (real covariates, synthetic autoregressive errors, exact target and exact conditional long-run covariance). The air-quality stream has an estimated variance inflation near 20, so dependence-blind intervals fail badly. (c) Coverage (vertical) against block length b (horizontal, log scale) on the traffic stream: the free diagnostic r_j of Section 4 falls below 0.05 exactly where coverage reaches nominal.

eight, which matters because those are the parameters a user would otherwise have to think about. (f) Gap and thinning behave as Proposition 10 predicts and do not move coverage, which is what the theory says: $\bar{H}$ enters the limit only through a rate. (g) At strong dependence ($\phi = \psi = 0.85$, $\kappa = 7.0$) the block length has to respect the dependence horizon: coverage improves monotonically as b grows from 20 to a few hundred, and the free diagnostic r_j falls in step with it, which is what makes it usable as a guide. (h) The weighted averaged variant is slightly better at small N and slightly worse at large N ; both are reported. The non-positive-definiteness fallback of Remark 4 never triggered (0.0% of all runs, all panels).

5.5 The correct interval is also the cheaper one

The authoritative statement is the operation count, because it is exact and independent of the machine. At fixed N and $b = m = d$, total work grows as $d^{1.56}$ for BGSN and as $d^{1.91}$ once the i.i.d. plug-in variance is added, because the plug-in score covariance costs $O(d^2)$ per *observation* whereas our long-run covariance costs $O(d^2)$ per *block* (Appendix B.5). BGSN’s exponent exceeds one at these settings, and the reason is (7): at $N = 200,000$ the one-off $O(c_w d^3)$ warm-up is comparable to the $O(dN)$ streaming pass once $d \geq 40$, so the measured exponent interpolates between 1 and 3. Evaluating the same exact counts at $N = 10^8$, where the warm-up is negligible, gives $d^{1.00}$ for BGSN against $d^{1.86}$ with the plug-in variance — the $O(dN)$ versus $O(d^2 N)$ separation the design predicts. Measured wall-clock times, on a shared and heavily loaded machine, give $d^{2.18}$ and $d^{1.90}$; they are noisier than the counts and we report them only as corroboration. In the main configuration BGSN runs in **[TBD]** times the time of the dependence-blind streaming Newton method — that is, our valid intervals cost less than the invalid ones, not more.

5.6 Real streams

?? reports two protocols on hourly traffic volume on Interstate 94 and hourly PM2.5 at twelve Beijing monitoring sites. After a fixed regression adjustment (weather plus calendar harmonics) the residual lag-one

autocorrelation is [TBD] and [TBD] respectively, and the estimated variance inflation is around 2 for traffic and around 20 for air quality. ?? reports, for each stream, the pooled prediction $d^{-1} \sum_j \{2\Phi(z_{0.025}/\sqrt{\kappa_j}) - 1\}$ next to the measurement, so the comparison is against the quantity the table actually measures rather than a single representative κ .

Protocol A keeps the real (strongly dependent) covariate stream and replaces the response by $x_t^\top \theta^* + u_t$ with u an autoregression whose coefficient equals that of the real residuals. This gives an *exact* target and an exact conditional long-run covariance while retaining real covariate dependence: BGSN covers [TBD] (traffic) and [TBD] (air quality) against nominal 0.95, while the dependence-blind interval covers [TBD] and [TBD]. Protocol B uses the real response and takes the full-stream M -estimator as the target; because that target is itself estimated from a superset of each segment, the correct nominal level for the comparison is $2\Phi(z_{0.025}/\sqrt{1 - 1/R}) - 1$, which we compute exactly and report in the table header. BGSN covers [TBD] and [TBD]. Neither protocol alone would settle the matter—A has an exact target but a synthetic error process, B has a real error process but an estimated target—and they agree.

Figure 7(c) shows why the block length is the one parameter a practitioner must think about on real data, and that the free diagnostic $r_j = |u_j^\top (\hat{S}^{\text{FT}} - \hat{S}^{\text{BM}}) u_j| / (u_j^\top \hat{S}^{\text{FT}} u_j)$ flags the problem: on the traffic stream r_j falls below 0.05 once $b \geq 40$, which is exactly where coverage reaches nominal.

6 Related work

This paper sits at the intersection of three literatures, and the honest statement of what is new requires being precise about each.

Streaming second-order methods with $O(dN)$ cost. Inversion-free stochastic Newton methods maintain a running inverse curvature estimate by rank-one Sherman–Morrison updates [80] and have been developed for logistic regression [6], Gauss–Newton problems [12], the geometric median [32], and general conditioned stochastic gradient descent [9, 51]; Chau et al. [13] survey the inversion-free idea more broadly, and Byrd et al. [11], Moritz et al. [61], Wang et al. [89] give the stochastic quasi-Newton and L-BFGS counterparts. Our starting point is Godichon-Baggioni and Werge [33], who thin the rank-one updates with a Bernoulli variable and take mini-batches of size d to reach $O(dN)$ total cost with asymptotic efficiency, and Godichon-Baggioni et al. [35], who reach the same budget by masking Hessian columns. **The $O(dN)$ budget and the idea of skipping curvature updates are theirs, not ours;** Chen et al. [15] independently thin Hessian *entries* with probability $O(1/d^2)$ in a zeroth-order method. All of this work assumes independent observations. Most of it proves central limit theorems and stops there, but not all: for averaged first-order algorithms under independent data Godichon-Baggioni [31] gives a recursive online estimator of the asymptotic variance with almost sure and L^2 rates, and Chen et al. [14], Godichon-Baggioni and Lu [32], Zhu et al. [99] supply plug-in and batch-means constructions. **Online estimation of the instantaneous sandwich under independent data is therefore not new either;** what is missing is the long-run object under dependence, and the streaming second-order setting. Our contribution on this axis is (i) the analysis of the rank-one recursion when the accumulated terms are dependent, (ii) replacing Bernoulli thinning by a deterministic gap, which is strictly better at matched cost (Proposition 10), and (iii) turning the central limit theorem into usable intervals.

Online inference for stochastic approximation. Under independent sampling this is a mature area. Chen et al. [14] give the plug-in and increasing-batch-means variance estimators for averaged stochastic gradient descent; Zhu et al. [99] make batch means fully recursive with growing overlapping windows and obtain $\|\hat{\Sigma} - \Sigma\| = O(N^{-1/8})$; Singh et al. [81] show equal batch sizes are preferable and apply

the flat-top bias correction; Ni and Huo [65] improve the batch-means rate to $N^{-1/6}$ and establish the minimax rate $\Theta(N^{-(1-a)/2})$ for *Hessian-free* covariance estimation from the iterate path; Wei et al. [90] reduce the bias by a cumulative-sum identity rather than a two-scale device. Fang [25], Fang et al. [26] bootstrap; Su and Zhu [82] split the trajectory; Chen et al. [15], Lee et al. [50], Li et al. [54] studentise by a random scaling matrix and estimate nothing. Leung and Chan [52] and Wu [96] design recursive long-run variance estimators directly. **None of these is a claim we duplicate**; what we take from them is the batch-means template and the flat-top correction, and what we add is the *block-gradient* construction and its consequence for the rate (Remark 3).

Two recent papers give online covariance estimation for streaming *second-order* methods: Kuang et al. [46] for a sketched Newton method and Wang et al. [88] for Nesterov-accelerated sketching. Both are under independent data, both target the instantaneous sandwich, and both require the gradient noise to be a martingale difference with respect to the algorithm’s filtration (46, Assumption 3.2) — exactly the condition that temporal dependence destroys. Kuang et al. [46] remark that their analysis extends to conditioned stochastic gradient methods generally, which would include an independent-data version of our algorithm; the martingale assumption is what keeps that remark from covering the present setting.

Stochastic approximation on dependent data. Convergence of stochastic approximation under Markovian or mixing noise is classical [4, 5, 7, 28, 42, 48], and modern rates are available for Markov-chain gradient descent [1, 19, 22, 23, 45, 62, 83]. Godichon-Baggioni et al. [34] analyse streaming stochastic gradient methods with time-dependent and biased gradients, and identify blocked or growing mini-batches as the device that breaks dependence; they give L^2 rates but no central limit theorem and no inference.

Blocked gradients as a dependence-breaking device are theirs.

On the inference side, Liu et al. [56] give a multiplier bootstrap for ϕ -mixing streams using alternating growing blocks, and establish the long-run sandwich limit and, qualitatively, the failure of dependence-blind intervals; Li et al. [53] give a functional central limit theorem and the semiparametric efficiency bound $H^{-1}SH^{-1}$ under Markovian sampling, with random scaling for inference; Samsonov et al. [79] give Berry–Esseen bounds and an overlapping-batch-means variance for *linear* stochastic approximation with Markovian noise; Roy and Balasubramanian [77] give an online batch-means estimator of the Markovian long-run sandwich; Li et al. [54] treat controlled Markov chains with random scaling. **The long-run sandwich limit, the efficiency bound, and the existence of online long-run covariance estimators under dependence are all prior art** — for first-order methods. Every one of these works is first-order; none maintains a curvature inverse, and the two that estimate a long-run covariance do so from the iterate path, at $O(N^{-1/8})$, with block lengths that must grow like $N^{3/4}$ (79, Prop. 2, Cor. 2). Remark 3 explains why block gradients escape that constraint, and Theorem 9 quantifies the gain.

Long-run variance estimation. The statistical device is heteroskedasticity- and autocorrelation-consistent covariance estimation [2, 3, 63, 64, 91, 93], its fixed- b alternative [43, 44, 84], and the batch-means literature from simulation output analysis and Markov chain Monte Carlo [16, 27, 29, 30, 41, 87]. The two-scale correction we use is the flat-top lag window of Politis and Romano [69], refined by Politis [68] and recast as “lugsail” lag windows by Vats and Flegal [86]; Singh et al. [81] bring it to stochastic gradient descent under independent data, where their Eqs. (15) and (17) coincide with our (11) and their Corollary 1 shows the convergence *rate* is unchanged. **We do not claim the correction.** We claim the bias order it attains at block-gradient resolution under geometric mixing, which is the property that makes $b \asymp \log N$ admissible, and the streaming implementation. Block bootstrap and subsampling [47, 49, 70] are the resampling counterparts.

Asymptotics for dependent M -estimation. The population statements we use are standard: sandwich asymptotics for M -estimators and generalised method of moments [37, 39, 85, 92], misspecified models with dependent observations [20], marginal models for correlated responses [55, 98] — the setting our logistic design imitates — and central limit theorems for mixing sequences [10, 18, 21, 38, 40, 57, 73, 76, 94]. The technical inputs to our proofs are Davydov’s covariance inequality [17, 72], the Bernstein-type inequality for weakly dependent sequences of Merlevède et al. [58], the almost sure behaviour of stochastic algorithms [24, 60, 66, 67, 75], and the functional central limit theorem of Herrndorf [38].

7 Limitations

We list the boundaries of the result plainly, because several of them are load-bearing.

Memory is $O(d^2)$, not $O(d)$. The $O(dN)$ claim is about *time*. The method stores a $d \times d$ inverse curvature estimate and two $d \times d$ covariance accumulators, so memory is $O(d^2)$; this is inherited from the independent-data method we extend and is not something we improve. For d in the thousands this is the binding constraint, and a sketched or diagonal-plus-low-rank curvature representation—as in Godichon-Baggioni et al. [35], Kuang et al. [46], Wang et al. [88]—would be the natural remedy. Whether the block-gradient long-run covariance estimator survives such a compression is open.

Geometric mixing is a real restriction, and the block length has to respect it. Theorem 9 buys $b \asymp \log N$ from the geometric decay in Lemma 11. Under algebraic mixing the flat-top bias is polynomial rather than exponential and the admissible b grows accordingly; long-memory streams are outside the theory altogether. This is not merely formal: our air-quality stream has a residual lag-one autocorrelation of 0.92 and a variance-inflation factor around 20, and there the block length needed for calibrated intervals is an order of magnitude larger than for the traffic stream (??). The free diagnostic $r_j = |u_j^\top (\widehat{S}^{\text{FT}} - \widehat{S}^{\text{BM}}) u_j| / (u_j^\top \widehat{S}^{\text{FT}} u_j)$ detects the problem—it estimates the relative block-length bias of plain batch means—but it does not remove it. It could be turned into an automatic rule by maintaining covariance accumulators at several block scales $b, 2b, 4b, \dots$ at once (one extra $O(d^2)$ update per block per scale, all from the same block gradients) and reporting from the smallest scale at which r_j falls below a tolerance; we did not implement this, and we do not claim an automatic rule for b .

$\widehat{S}^{\text{FT}}$ need not be positive semidefinite. This is intrinsic to flat-top and lugsail estimators [81, Remark 5]. Marginal intervals need only positive scalars $u_j^\top \widehat{S}^{\text{FT}} u_j$, and the documented fallback to $\widehat{S}^{\text{BM}}$ never triggered in any of our runs, but a user who needs the full matrix (for a joint test, say) must project it.

The curvature form is required. Everything rests on $\nabla^2 F(\theta) = \mathbb{E}[\alpha \Psi \Psi^\top]$, which holds for generalised linear models, ridge and softmax regression, the geometric median and Gauss–Newton problems, but not for a general objective. Without it there is no rank-one recursion and no inversion-free update.

Warm-up is a real requirement, not an implementation detail. The $1/t$ schedule with a poor preconditioner leaves a slowly decaying component that dominates the sampling error at realistic N : with a negligible warm-up, coverage of nominal 95% intervals fell to [TBD] in our autoregressive design even though the variance estimate stayed accurate (??(c)). We give a scaling rule ($c_w d$ curvature updates, with $c_w = 50$ throughout) and show insensitivity above it, but this is a tuning constant, and the asymptotic theory is silent about it. It also has a cost consequence we state rather than bury: the warm-up performs $c_w d$

rank-one updates at $O(d^2)$ each, so the total time is $O(dN + c_w d^3)$ and the advertised $O(dN)$ only bites once $N \gtrsim c_w d^2$ (with $c_w = 50$ and $d = 80$ that means $N \gtrsim 3 \times 10^5$; at the $N = 2 \times 10^5$ of Appendix B.5 the warm-up is comparable to the streaming pass).

Rates and what we do not prove. We give an almost sure rate and a central limit theorem, not a Berry–Esseen bound; the first-order Markovian literature has the latter for linear stochastic approximation [79], and we do not match it. We attain the semiparametric efficiency bound of Li et al. [53] rather than establishing it. Our $\tilde{O}(N^{-1/2})$ rate for $\widehat{S}^{\text{FT}}$ is not comparable to the minimax rate for Hessian-free covariance estimation from an iterate path [65], because we use more information (block gradients and a maintained curvature estimate); we make no claim about optimality within our own information set.

Scope of the empirical evidence. All designs are convex M -estimation with $d \leq 160$ and $N \leq 4 \times 10^5$; there is no deep-learning experiment and the theory would not support one. The dependence in the simulated designs is stationary and geometrically decaying by construction. On the real streams we do not know the truth, which is why we run two protocols; the semi-synthetic protocol has an exact target but a synthetic (autoregressive) error process, and the fully real protocol has a real error process but a target that is itself estimated—we correct the nominal level for that exactly, but the correction assumes the segments are approximately exchangeable, which non-stationarity would violate. Neither protocol alone would be convincing; together they bracket the honest answer.

One thing we deliberately do not claim. Gapping improves the curvature estimate at matched cost (Proposition 10, Figure 4), but the curvature estimate enters the limit distribution only through a rate, so the end-to-end effect on the point estimate and on coverage is second order and we could not measure it reliably. The case for gapping is that it is free, deterministic, and provably better than the randomised alternative—not that it changes the headline numbers.

8 Conclusion

On a dependent stream, a streaming quasi-Newton method’s point estimate can be right while its stated uncertainty is wrong by a fixed factor, and the fix is not an add-on: blocking the gradient step, which the $O(dN)$ budget already suggests, is exactly what makes the long-run covariance estimable, and gapping the curvature updates is what makes the inversion-free recursion behave as it would on independent data. The resulting method costs less than the dependence-blind alternative it replaces, because a plug-in instantaneous score covariance costs $O(d^2)$ per observation whereas a blocked long-run covariance costs $O(d^2)$ per block.

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A Proofs

Throughout, C denotes a finite constant that may change from line to line, $\delta_t = \theta_t - \theta^\star$, $\varepsilon_t = \bar{g}_t(\theta^\star)$, $\gamma_t = 1/t$, $P_t = \bar{H}_t^{-1}$, and $\Lambda_k = \mathbb{E}[\varepsilon_0 \varepsilon_k^\top]$. We write n_t for the number of rank-one curvature updates completed by the end of block t ; with gap m and no Bernoulli thinning, $n_t = \lfloor tb/m \rfloor$, so $n_t \asymp t$.

Order of the argument. The results are proved in the order of Remark 2: Lemma 1 (from the ridge alone, no assumption on the iterates); Theorem 3 (uses Lemma 1 only); the preliminary rate $\|\delta_t\| = O(t^{-1/2} \log^c t)$ (uses Theorem 3 and consistency of $\bar{H}$, not its rate); Theorem 2; and then Theorems 4, 5, 8 and 9. Every remainder bound below is uniform over $b \leq C \log N$, which is the only regime in which Theorem 9 uses them, and the b -dependence is displayed wherever it is not $O(1)$.

A.1 Preliminaries on dependence

Lemma 11 (Geometric decay of the score autocovariances). *Under Assumption 5, $\|\Gamma_k\|_{\text{op}} \leq C\rho^{|k|}$ with $\rho = c_{\text{mix}}^{\nu/(2+\nu)}$, and $\|\Lambda_k\|_{\text{op}} \leq Cb^{-1}\rho^{(|k|-1)b}$ for $|k| \geq 1$ while $\|\Lambda_0\|_{\text{op}} \leq C/b$. Consequently $\sum_{k \in \mathbb{Z}} \|\Lambda_k\|_{\text{op}} \leq C/b$ and $S = \sum_k \Gamma_k$ converges absolutely.*

Proof. Davydov's inequality [17], in the form of Rio [72, Cor. 1.1], gives for random variables $U \in \sigma(\xi_i, i \leq 0)$, $V \in \sigma(\xi_i, i \geq k)$ with $2 + \nu$ moments, $|\text{Cov}(U, V)| \leq 8 \alpha_{\text{mix}}(k)^{\nu/(2+\nu)} \|U\|_{2+\nu} \|V\|_{2+\nu}$. Applying it coordinatewise to g_0 and g_k and using Assumption 5 gives the first claim. For the second, $\Lambda_k = b^{-2} \sum_{i \in B_0} \sum_{j \in B_k} \Gamma_{j-i}$; for $k \geq 1$ every lag $j-i$ is at least $(k-1)b+1$, so $\|\Lambda_k\|_{\text{op}} \leq b^{-2} \cdot b^2 \cdot C\rho^{(k-1)b}$, and for $k = 0$, $\|\Lambda_0\|_{\text{op}} \leq b^{-2} \sum_{|l| < b} (b - |l|) C\rho^{|l|} \leq Cb^{-1}$. Summing the geometric series gives $\sum_k \|\Lambda_k\|_{\text{op}} \leq C/b$. $\square$

Lemma 12 (Martingale–coboundary decomposition). *Let $(\zeta_t)_{t \geq 1}$ be stationary, centred, $\mathbb{R}^q$ -valued, $\mathcal{F}_t$ -measurable with $\mathcal{F}_t = \sigma(\xi_i : i \leq tb)$, and let $\mathbb{E}\|\zeta_1\|^{2+\nu} < \infty$ under Assumption 5. Then $\sum_{n \geq 0} \|\mathbb{E}[\zeta_n | \mathcal{F}_0]\|_2 < \infty$, the series $z_t = \sum_{n \geq 0} \mathbb{E}[\zeta_{t+n} | \mathcal{F}_t]$ converges in L^2 , and*

$$\zeta_t = m_{t+1} + z_t - z_{t+1}, \quad m_{t+1} = z_{t+1} - \mathbb{E}[z_{t+1} | \mathcal{F}_t], \quad (12)$$

where (m_t) is a stationary martingale difference sequence with respect to $(\mathcal{F}_t)$ and $\mathbb{E}\|m_1\|^2 < \infty$, $\mathbb{E}\|z_1\|^2 < \infty$. Moreover $\mathbb{E}[m_1 m_1^\top] = b^{-1}S$ when $\zeta_t = \varepsilon_t$.

Proof. For U that is $\sigma(\xi_i : i \geq n)$ -measurable with $2 + \nu$ moments, the standard consequence of strong mixing (see Rio 73, Ch. 1) is $\|\mathbb{E}[U | \mathcal{F}_0]\|_2 \leq C \alpha_{\text{mix}}(n)^{\nu/(2(2+\nu))} \|U\|_{2+\nu}$, which under Assumption 5 decays geometrically in n and is therefore summable; this is Gordin's condition, and (12) together with the stated moment bounds is Gordin's decomposition; see Hall and Heyde [36, Ch. 5]. Summing (12) telescopes the coboundary, so $\sum_{t \leq T} \zeta_t = \sum_{t \leq T} m_{t+1} + z_1 - z_{T+1}$; dividing by $\sqrt{T}$ and using $z_{T+1}/\sqrt{T} \rightarrow 0$ in probability identifies $\lim T^{-1} \text{Var}(\sum_{t \leq T} \zeta_t) = \mathbb{E}[m_1 m_1^\top]$, which for $\zeta_t = \varepsilon_t$ is $b^{-1}S$ by (1). $\square$

Lemma 13 (Weighted sums with adapted weights: convergence and rate). *Let (ζ_t) be as in Lemma 12, let (A_t) be $\mathcal{F}_{t-1}$ -measurable random matrices, and let (γ_t) be deterministic and non-increasing. Suppose there are deterministic (c_t) , (u_t) and a finite random Λ with, almost surely for all t ,*

$$\|A_t\|_{\text{op}} \leq \Lambda c_t \quad \text{and} \quad \|\gamma_{t+1} A_{t+1} - \gamma_t A_t\|_{\text{op}} \leq \Lambda u_t, \quad (13)$$

and write $V_T = \sum_{t \leq T} \gamma_t^2 c_t^2$ and $U_T = \sum_{t \leq T} u_t$. Then for every $\varpi > 0$, almost surely,

$$\left\| \sum_{t \leq T} \gamma_t A_t \zeta_t \right\| = O(V_T^{1/2} \log^{1/2+\varpi} T) + O(U_T \log^{1+\varpi} T) + O(\gamma_T c_T T^{1/2+\varpi}). \quad (14)$$

If moreover $V_\infty < \infty$, $U_\infty < \infty$ and $\gamma_T c_T T^{1/2+\varpi} \rightarrow 0$ for some $\varpi > 0$, then $\sum_t \gamma_t A_t \zeta_t$ converges almost surely.

Proof. Localisation. For $K \in \mathbb{N}$ replace A_t by $A_t \mathbf{1}\{\sup_{s \leq t} \max(\|A_s\|_{\text{op}}/c_s, \|\gamma_{s+1}A_{s+1} - \gamma_s A_s\|_{\text{op}}/u_s) \leq K\}$, which is still $\mathcal{F}_{t-1}$ -measurable (the indicator is decreasing in t and $\mathcal{F}_{t-1}$ -measurable at each t because it involves A_s only for $s \leq t$, and A_{s+1} enters only through $s+1 \leq t$). On $\{\Lambda \leq K\}$ the two processes agree and $\mathbb{P}(\bigcup_K \{\Lambda \leq K\}) = 1$, so it suffices to prove the claims under the *deterministic* envelopes $\|A_t\|_{\text{op}} \leq Kc_t$, $\|\gamma_{t+1}A_{t+1} - \gamma_t A_t\|_{\text{op}} \leq Ku_t$. Fix K and drop it.

Martingale part. Insert (12): $M_T = \sum_{t \leq T} \gamma_t A_t m_{t+1}$ is a martingale, because A_t is $\mathcal{F}_t$ -measurable and $\mathbb{E}[m_{t+1} | \mathcal{F}_t] = 0$ — this is the exact cancellation the decomposition buys, and it holds whatever A_t does. Its predictable quadratic variation obeys

$$\langle M \rangle_T = \sum_{t \leq T} \gamma_t^2 \mathbb{E}[\|A_t m_{t+1}\|^2 | \mathcal{F}_t] \leq K^2 \sum_{t \leq T} \gamma_t^2 c_t^2 v_t, \quad v_t := \mathbb{E}[\|m_{t+1}\|^2 | \mathcal{F}_t],$$

with (v_t) stationary, non-negative and $\mathbb{E}v_t = \mathbb{E}\|m_1\|^2 < \infty$. By Tonelli $\mathbb{E}[\sum_{t \leq T} \gamma_t^2 c_t^2 v_t] = \mathbb{E}\|m_1\|^2 V_T$, so Markov's inequality along $T = 2^j$ and Borel–Cantelli give $\langle M \rangle_T = O(V_T \log^\varpi T)$ a.s. (and $\langle M \rangle_\infty < \infty$ a.s. if $V_\infty < \infty$). The martingale law of the iterated logarithm then gives $\|M_T\| = O(V_T^{1/2} \log^{1/2+\varpi} T)$ a.s., and if $\langle M \rangle_\infty < \infty$ a locally square-integrable martingale converges a.s.

Coboundary part. Abel summation gives

$$\sum_{t \leq T} \gamma_t A_t (z_t - z_{t+1}) = \gamma_1 A_1 z_1 - \gamma_T A_T z_{T+1} + \sum_{t < T} (\gamma_{t+1} A_{t+1} - \gamma_t A_t) z_{t+1}.$$

The last sum is at most $K \sum_{t < T} u_t \|z_{t+1}\|$, a non-negative sum with expectation $K\mathbb{E}\|z_1\|U_T$, hence $O(U_T \log^{1+\varpi} T)$ a.s. by Markov and Borel–Cantelli along dyadic blocks; if $U_\infty < \infty$ its expectation is finite and it converges a.s. Finally $\|\gamma_T A_T z_{T+1}\| \leq K\gamma_T c_T \|z_{T+1}\| = O(\gamma_T c_T T^{1/2+\varpi})$ a.s., because $\mathbb{E}\|z_1\|^2 < \infty$ and Borel–Cantelli give $\|z_T\| = o(T^{1/2+\varpi})$. $\square$

Remark 5 (Why the lemma is stated this way). Three features are load-bearing and each corresponds to a place where a naive argument fails. (i) The weights are *random and adapted*: they contain $\bar{H}_{t-1}^{-1}$, so a lemma for deterministic weights would not apply. Note that adaptedness is essential and not cosmetic: for bounded adapted weights and a mixing, centred, stationary ζ the sum $\sum_t a_t \zeta_t$ can grow like T if a_t is allowed to correlate with ζ_t — take $\zeta_t = u_t + u_{t-1}$ with u_t i.i.d. signs and $a_t = \zeta_{t-1}$ — which is exactly the non-martingale obstruction. The decomposition (12) removes it because $\mathbb{E}[m_{t+1} | \mathcal{F}_t] = 0$ *exactly*, whatever A_t does. (ii) The envelope is deterministic only up to a finite random factor, which is all that is available for weights built from the iterates; the localisation handles it. (iii) We use a martingale law of the iterated logarithm rather than a Bernstein inequality for mixing sums. The Bernstein inequality of Merlevède et al. [58] is stated for uniformly *bounded* summands, which Assumption 5 does not provide; after (12) the summands form a martingale difference sequence and no such inequality is needed. (iv) The three terms of (14) are needed separately: the applications below use $\gamma_t \equiv 1$, for which U_T can diverge logarithmically, so a hypothesis $U_\infty < \infty$ would be too strong.

Lemmas 12 and 13 are the technical substitute for the martingale argument used in the independent-data analysis. Two features matter. First, the weights may be *random and adapted*, which is what we need because they involve $\bar{H}_{t-1}^{-1}$. Second, the conditions (13) are of the form already familiar from the independent-data theory. The vanishing ridge supplies the deterministic envelope: it forces $\lambda_{\min}(A_n) \geq c n^{1-q_r}$, hence $\|\bar{H}_t^{-1}\|_{\text{op}} = W_n \|A_n^{-1}\|_{\text{op}} \leq c^{-1} n^{q_r} =: c_t$ with $q_r < 1/2$, so with $\gamma_t = 1/t$ the first half of (13) is $\sum_t O(t^{2q_r-2}) < \infty$. The second half holds because a single rank-one Riccati update changes the preconditioner by only $O(1/t)$:

Lemma 14 (Increments of the preconditioner). *Under the assumptions of Theorem 2, $\|\bar{H}_t^{-1} - \bar{H}_{t-1}^{-1}\|_{\text{op}} = O(1/t)$ a.s., hence with $\gamma_t = 1/t$ and $A_t = \bar{H}_{t-1}^{-1}$ we have $\|\gamma_{t+1}A_{t+1} - \gamma_t A_t\|_{\text{op}} = O(t^{-2})$ a.s. and the second half of (13) holds for every $\delta < 1/2$.*

Proof. $\bar{H}_t^{-1} = W_{n_t} A_{n_t}^{-1}$ and, from block $t-1$ to block t , W increases by $O(1)$ and A^{-1} changes by $O(1)$ rank-one Sherman–Morrison steps $A^{-1} \mapsto A^{-1} - c A^{-1} u u^\top A^{-1} / (1 + c u^\top A^{-1} u)$, each of operator norm at most $\|A^{-1}\|_{\text{op}}$ times $c \|A^{-1}\|_{\text{op}} \|u\|^2 / (1 + c u^\top A^{-1} u) \leq 1$, and in fact at most $\|A^{-1}\|_{\text{op}}^2 c \|u\|^2 = O(t^{-2})$ once $\lambda_{\min}(A_{n_t}) \geq c't$ (which holds a.s. for large t because $A_{n_t}/n_t \rightarrow H \succ 0$). Multiplying by $W_{n_t} = O(t)$ gives $O(1/t)$; the extra factor $\gamma_t = 1/t$ and $\gamma_{t+1} - \gamma_t = O(t^{-2})$ give the second claim. $\square$

A.2 The ridge bound (Lemma 1)

Each curvature update adds $\iota_n e_{k_n} e_{k_n}^\top \succeq 0$ along a cycling basis direction, so after n updates every basis direction has received at least $\lfloor n/d \rfloor$ of them and $A_n \succeq \lambda_0 \zeta_0 I + (d^{-1} \sum_{k \leq n} \iota_k) I$. With $\iota_k = c_\iota \zeta_k k^{-q_r}$ and $\zeta_k \geq \zeta_* > 0$ almost surely (finite because $\zeta_k = d/\text{tr}(\bar{H}_k^{-1})$ and $\bar{H}_k$ is bounded above, being an average of terms with $\eta' > 2$ moments), $\sum_{k \leq n} \iota_k \geq c n^{1-q_r}$, hence $\lambda_{\min}(A_n) \geq c' n^{1-q_r}$ and $\|\bar{H}_t^{-1}\|_{\text{op}} = W_{n_t} \|A_{n_t}^{-1}\|_{\text{op}} \leq (1 + n_t)/(c' n_t^{1-q_r}) = O(n_t^{q_r})$. The random constant is $c_1 = c' \zeta_*$; nothing about the iterates is used. $\square$

A.3 The curvature rate (Theorem 2)

Write $Y_n = \alpha(\theta^*; \xi_{i_n}) \Psi(\theta^*; \xi_{i_n}) \Psi(\theta^*; \xi_{i_n})^\top$ and $\tilde{Y}_n = \alpha(\theta_{t(n)-1}; \xi_{i_n}) \Psi \Psi^\top$ for the term actually accumulated, where $t(n)$ is the block containing i_n . Then, by (4) and (5),

$$\bar{H}_t - H = \underbrace{\frac{\lambda_0 \zeta_0 I}{W_{n_t}}}_{(i)} + \underbrace{\frac{1}{W_{n_t}} \sum_{n \leq n_t} \iota_n e_{k_n} e_{k_n}^\top}_{(ii)} + \underbrace{\frac{1}{W_{n_t}} \sum_{n \leq n_t} (Y_n - H)}_{(iii)} + \underbrace{\frac{1}{W_{n_t}} \sum_{n \leq n_t} (\tilde{Y}_n - Y_n)}_{(iv)}. \quad (15)$$

Since $W_{n_t} = 1 + n_t \asymp t$, term (i) is $O(1/t)$. For (ii), $\|\cdot\|_{\text{op}} \leq n_t^{-1} \sum_{n \leq n_t} \iota_n \leq C n_t^{-q_r}$ because $\iota_n \leq C n^{-q_r}$ and $q_r < 1$; the same bound from below along each basis direction gives, together with $H \succ 0$, $\lambda_{\min}(\bar{H}_t) \geq c n_t^{-q_r}$ for all t (this is the only place the ridge is used, and it is what bounds $\|\bar{H}_t^{-1}\|_{\text{op}}$ for every t rather than eventually). Term (iv) is bounded, by the local Lipschitz property in Assumption 3 and Cauchy–Schwarz, by $C n_t^{-1} \sum_{n \leq n_t} \|\theta_{t(n)-1} - \theta^*\| \cdot \|\Psi \Psi^\top\|_{\text{op}}$, which under the assumed preliminary rate is $O(n_t^{-1} \sum_{n \leq n_t} n^{-1/2} \log^c n) = O(n_t^{-1/2} \log^c n_t)$ a.s.

The new content is term (iii). The array $(Y_n)_{n \geq 1}$ is the subsequence of a stationary geometrically mixing sequence at spacing m , hence itself stationary with mixing coefficients $\alpha_{\text{mix}}(jm) \leq K c_{\text{mix}}^{jm}$, and has $\eta' > 2$ moments by Assumption 3. Applying Lemma 13 with $a_n \equiv 1$ (i.e. its maximal-inequality half, which does not need square-summability) gives $\|\sum_{n \leq n_t} (Y_n - H)\|_{\text{op}} = O(n_t^{1/2} \log^{1/2+\varpi} n_t)$ a.s., so (iii) = $O(t^{-1/2} \log^{1/2+\varpi} t)$ a.s. Collecting the four terms, $\|\bar{H}_t - H\|_{\text{op}} = O(t^{-\min\{q_r, 1/2\}} \log^{c'} t)$ a.s., which is $O(t^{-\nu_H})$ for every $\nu_H < \min\{q_r, 1/2\}$. Finally, on the event $\|\bar{H}_t - H\|_{\text{op}} \leq \lambda_{\min}(H)/2$ (which holds for all large t a.s.), $\|\bar{H}_t^{-1} - H^{-1}\|_{\text{op}} \leq \|\bar{H}_t^{-1}\|_{\text{op}} \|\bar{H}_t - H\|_{\text{op}} \|H^{-1}\|_{\text{op}} \leq 2 \lambda_{\min}(H)^{-2} \|\bar{H}_t - H\|_{\text{op}}$. $\square$

A.4 Consistency (Theorem 3)

Substituting $\bar{g}_t(\theta_{t-1}) = \nabla F(\theta_{t-1}) + \zeta_t$ into (6), where $\zeta_t = \bar{g}_t(\theta_{t-1}) - \nabla F(\theta_{t-1})$, and expanding $F(\theta_t) - F(\theta^*)$ to second order using Assumption 1,

$$\begin{aligned} F(\theta_t) - F(\theta^*) &\leq F(\theta_{t-1}) - F(\theta^*) - \gamma_t \nabla F(\theta_{t-1})^\top P_{t-1} \nabla F(\theta_{t-1}) \\ &\quad - \gamma_t \nabla F(\theta_{t-1})^\top P_{t-1} \zeta_t + \frac{L}{2} \gamma_t^2 \|P_{t-1}\|_{\text{op}}^2 \|\bar{g}_t(\theta_{t-1})\|^2. \end{aligned} \quad (16)$$

The last term is summable: $\gamma_t^2 \|P_{t-1}\|_{\text{op}}^2 = O(t^{2q_r-2})$ by Theorem 2 with $q_r < 1/2$, and $\mathbb{E}\|\bar{g}_t(\theta_{t-1})\|^2 \leq C + C'(F(\theta_{t-1}) - F(\theta^*))$ by Assumption 2 and Jensen. The cross term is the one that would be handled by a martingale argument under independence. Instead, decompose $\zeta_t = \varepsilon_t + r_t$ with $r_t = \bar{g}_t(\theta_{t-1}) - \nabla F(\theta_{t-1}) - \varepsilon_t$, so that $\|r_t\| \leq \|\hat{H}_t - H\|_{\text{op}} \|\delta_{t-1}\| + L_\delta \|\delta_{t-1}\|^2$ where $\hat{H}_t$ is the block curvature; r_t is dominated by $\|\delta_{t-1}\|$ times an L^2 -bounded factor, and its contribution to (16) is absorbed by the Robbins–Siegmund argument [75]. The remaining term $-\sum_t \gamma_t \nabla F(\theta_{t-1})^\top P_{t-1} \varepsilon_t$ is handled by Lemma 13 with $A_t = \nabla F(\theta_{t-1})^\top P_{t-1}$, which is $\mathcal{F}_{t-1}$ -measurable as required. Its first condition holds because $\|P_{t-1}\|_{\text{op}} = O(t^{q_r})$ with $q_r < 1/2$ (Theorem 2) and $\mathbb{E}\|\nabla F(\theta_{t-1})\|^2$ is bounded on the events $\{\sup_{s \leq t} \|\theta_s\| \leq K\}$, over which we localise and then let $K \rightarrow \infty$; its second condition holds by Lemma 14 together with $\|\nabla F(\theta_t) - \nabla F(\theta_{t-1})\|_{\text{op}} \leq L\|\theta_t - \theta_{t-1}\| = O(\gamma_t t^{q_r})$. Robbins–Siegmund then gives $F(\theta_t) \rightarrow F(\theta^*)$ a.s. and $\sum_t \gamma_t \nabla F(\theta_{t-1})^\top P_{t-1} \nabla F(\theta_{t-1}) < \infty$ a.s.; and since $\gamma_t \lambda_{\min}(P_{t-1}) \geq \gamma_t / \|\bar{H}_{t-1}\|_{\text{op}} \geq c t^{-1}$ (because $\|\bar{H}_t\|_{\text{op}}$ is bounded above a.s., $\bar{H}_t$ being an average of terms with $\eta' > 2$ moments) we have $\sum_t \gamma_t \lambda_{\min}(P_{t-1}) = \infty$; Assumption 4 then forces $\theta_t \rightarrow \theta^*$ a.s. $\square$

A.5 Proof of Theorem 4 (rate) and Proposition 6 and Theorem 5 (CLT)

Write (6) as $\delta_t = \delta_{t-1} - \gamma_t P_{t-1} \{H\delta_{t-1} + \varepsilon_t + r_t\}$ with r_t as above. Multiplying by t and using $\gamma_t = 1/t$,

$$t \delta_t = (t-1)\delta_{t-1} + (I - P_{t-1}H)\delta_{t-1} - P_{t-1}\varepsilon_t - P_{t-1}r_t, \quad (17)$$

so that, summing from $t = 1$ to T (Abel summation) and dividing by T ,

$$\delta_T = -\frac{1}{T}H^{-1} \sum_{t \leq T} \varepsilon_t + \underbrace{\frac{1}{T} \sum_{t \leq T} (H^{-1} - P_{t-1})\varepsilon_t}_{(A)} + \underbrace{\frac{1}{T} \sum_{t \leq T} (I - P_{t-1}H)\delta_{t-1}}_{(B)} - \underbrace{\frac{1}{T} \sum_{t \leq T} P_{t-1}r_t}_{(C)}. \quad (18)$$

The leading term is *exactly* $-N_T^{-1}H^{-1} \sum_{i \leq N_T} g_i$, because $\sum_{t \leq T} \varepsilon_t = b^{-1} \sum_{i \leq N_T} g_i$ and $N_T = bT$. This identity is Proposition 6: the leading term does not depend on b .

Term (A). Apply (14) with $\gamma_t \equiv 1$, $\zeta_t = \varepsilon_t$ and the $\mathcal{F}_{t-1}$ -measurable weights $A_t = H^{-1} - P_{t-1}$: by Theorem 2 the envelope is $c_t = t^{-\nu_H}$, and by Lemma 14 the increments are $u_t = O(t^{-1})$. Hence $V_T = O(T^{1-2\nu_H})$ (as $\nu_H < 1/2$), $U_T = O(\log T)$, and $\gamma_T c_T T^{1/2+\varpi} = O(T^{1/2-\nu_H+\varpi})$, so

$$\left\| \sum_{t \leq T} (H^{-1} - P_{t-1})\varepsilon_t \right\| = O(T^{1/2-\nu_H} \log^{1/2+\varpi} T) + O(\log^{2+\varpi} T) + O(T^{1/2-\nu_H+\varpi}) = O(T^{1/2-\nu_H+\varpi}) \text{ a.s.}$$

Choosing $\varpi < \nu_H$ gives (A) = $O(T^{1/2-\nu_H+\varpi}) = o(T^{-1/2})$ a.s. It is worth seeing what does the work. The only bound available without (12) is $\|\mathbb{E}[(H^{-1} - P_{t-1})\varepsilon_t]\| \leq \|H^{-1} - P_{t-1}\|_{\text{op}} \|\varepsilon_t\|_2 = O(t^{-\nu_H} b^{-1/2})$ — Davydov gives no decay across a one-block gap, since the weight and the summand are only one block apart — and averaging that over $t \leq T$ leaves $O(b^{-1/2} T^{-\nu_H})$, which is *not* $o(T^{-1/2})$ for any admissible $\nu_H < 1/2$. The extra $T^{-1/2}$ comes entirely from the exact cancellation $\mathbb{E}[A_t m_{t+1} \mid \mathcal{F}_t] = 0$.

Term (B). $\|(B)\| \leq T^{-1} \sum_{t \leq T} \|I - P_{t-1}H\|_{\text{op}} \|\delta_{t-1}\| = T^{-1} \sum_{t \leq T} O(t^{-\nu_H}) \|\delta_{t-1}\|$, which is a pathwise bound needing no lemma. With the rung-(iii) preliminary rate $\|\delta_t\| = O(t^{-1/2} \log^c t)$ this is $O(T^{-1} \cdot T^{1/2-\nu_H} \log^c T) = o(T^{-1/2})$ a.s.

Term (C). $r_t = (\hat{H}_t - H)\delta_{t-1} + O(\|\delta_{t-1}\|^2)$, where $\hat{H}_t$ is the block curvature. The quadratic part contributes $O(T^{-1} \sum_t \log t/t) = O(\log^2 T/T) = o(T^{-1/2})$. For the first part take $\zeta_t = \hat{H}_t - H$, which is stationary, centred and square integrable under Assumption 3 (it is the block average of the terms appearing in (15) evaluated at θ^* , plus a Lipschitz remainder in θ_{t-1} bounded exactly as term (iv) there), and apply

(14) with $\gamma_t \equiv 1$ and the $\mathcal{F}_{t-1}$ -measurable weights acting by $A_t \zeta_t = P_{t-1} \zeta_t \delta_{t-1}$. By Lemma 1 and the rung-(iii) rate the envelope is $c_t = O(t^{q_r} \cdot t^{-1/2} \log^c t)$ and the increments are $u_t = O(t^{-1} c_t)$, so

$$V_T = O\left(\sum_{t \leq T} t^{2q_r-1} \log^{2c} t\right) = O(T^{2q_r} \log^{2c} T), \quad U_T = O\left(\sum_t t^{q_r-3/2} \log^c t\right) = O(1) \quad (q_r < 1/2),$$

and $\gamma_T c_T T^{1/2+\varpi} = O(T^{q_r+\varpi} \log^c T)$. Hence the sum is $O(T^{q_r+\varpi} \log^{c+1/2+\varpi} T)$ and (C) = $O(T^{q_r-1+\varpi} \log^{c'} T) = o(T^{-1/2})$ a.s. because $q_r < 1/2$. No independence between $\hat{H}_t$ and δ_{t-1} is used or needed: adjacent blocks are only $O(1/b)$ apart in the sense of Lemma 11, not $O(\rho^b)$, and an earlier version of this proof wrongly appealed to the latter.

Substituting into (18) gives $\|\delta_T\|^2 = O(\log N_T/N_T)$ a.s. (Theorem 4) and $\sqrt{N_T} \delta_T = -N_T^{-1/2} H^{-1} \sum_{i \leq N_T} g_i + o_p(1)$. The central limit theorem for strongly mixing stationary sequences with geometric coefficients and $2 + \nu$ moments [38, 40] gives $N^{-1/2} \sum_{i \leq N} g_i \rightsquigarrow \mathcal{N}(0, S)$, and Slutsky completes Theorem 5. The averaged variant follows from the same decomposition with $\gamma_t = ct^{-a}$ and the standard Abel argument for weighted averages [60, 67]. $\square$

A.6 The L^2 rate (Theorem 8)

Take expectations in (18). The leading term has $\mathbb{E}\|N^{-1} H^{-1} \sum_{i \leq N} g_i\|^2 = \text{tr}(H^{-1} S H^{-1})/N + o(1/N)$ by Lemma 11, uniformly in b because it does not involve b at all. For the remainders, replace each almost sure bound above by its L^2 counterpart: the martingale parts are bounded in L^2 by their predictable variations, which were already computed in expectation in the proof of Lemma 13, and the coboundary parts by $\mathbb{E}\|z_1\| \sum_t u_t$. On the descent side, iterating (16) in expectation with $\mathbb{E}\|\bar{g}_t(\theta_{t-1})\|^2 \leq C + C' \mathbb{E}[F(\theta_{t-1}) - F(\theta^*)]$ (Assumption 2) and $\gamma_t^2 \|P_{t-1}\|_{\text{op}}^2 = O(t^{2q_r-2})$ (Lemma 1) gives $\mathbb{E}[F(\theta_t) - F(\theta^*)] = O(\log t/t)$ by the standard Chung-type recursion, and local strong convexity converts this to $\mathbb{E}\|\delta_t\|^2 = O(\log t/t) = O(\log N_t/N_t) \cdot b$; the extra factor b is absorbed because $N_t = bt$ and the leading term is b -free. All constants depend on b only through $\mathbb{E}\|\bar{g}_t\|^2 \leq \mathbb{E}\|g_1\|^2$, which is monotone in b , so the bound is uniform over $b \leq C \log N$. $\square$

A.7 Coverage of dependence-blind intervals (Proposition 7)

By Theorem 5 and Theorem 2, $\sqrt{N}(\theta_{T,j} - \theta_j^*) \rightsquigarrow \mathcal{N}(0, V_{jj})$ with $V = H^{-1} S H^{-1}$, while $[\bar{H}^{-1} \hat{\Gamma}_0 \bar{H}^{-1}]_{jj} \rightarrow_p [H^{-1} \Gamma_0 H^{-1}]_{jj} = V_{jj}/\kappa_j$. Hence the studentised statistic converges to $\mathcal{N}(0, \kappa_j)$ and $\mathbb{P}(|\mathcal{N}(0, \kappa_j)| \leq z_{\alpha/2}) = 2\Phi(z_{\alpha/2}/\sqrt{\kappa_j}) - 1$. For the homogeneous AR design, $\Gamma_k = \sigma_u^2 (\phi\psi)^{|k|} \Sigma_x$ gives $S = \sigma_u^2 \Sigma_x \sum_k (\phi\psi)^{|k|} = \sigma_u^2 \Sigma_x (1 + \phi\psi)/(1 - \phi\psi)$ and $H = \Gamma_0/\sigma_u^2 = \Sigma_x$, so $V = \kappa \sigma_u^2 \Sigma_x^{-1}$ with $\kappa = (1 + \phi\psi)/(1 - \phi\psi)$ and the inflation is the same in every coordinate. $\square$

A.8 Long-run covariance estimation (Theorem 9)

(i) *Bias of batch means.* For the idealised estimator built from $\varepsilon_t = \bar{g}_t(\theta^*)$, stationarity gives $\mathbb{E}[b \varepsilon_t \varepsilon_t^\top] = b\Lambda_0 = \sum_{|k| < b} (1 - |k|/b) \Gamma_k$. Splitting,

$$b\Lambda_0 = S - \sum_{|k| \geq b} \Gamma_k - \frac{1}{b} \left(\Delta - \sum_{|k| \geq b} |k| \Gamma_k \right), \quad \Delta = \sum_{k \geq 1} k(\Gamma_k + \Gamma_k^\top), \quad (19)$$

and by Lemma 11 both remainder sums are $O(b\rho^b)$, giving $\mathbb{E}[\hat{S}^{\text{BM}}] = S - \Delta/b + O(\rho^b)$ up to the drift term treated below.

(ii) *Bias of the two-scale estimator.* With $\Lambda_1 = \mathbb{E}[\varepsilon_0 \varepsilon_1^\top] = b^{-2} \sum_{i \in B_0, j \in B_1} \Gamma_{j-i}$ and counting multiplicities of each lag (k for $1 \leq k \leq b$, $2b - k$ for $b < k < 2b$),

$$b(\Lambda_1 + \Lambda_1^\top) = \sum_{1 \leq |k| \leq b} \frac{|k|}{b} \Gamma_k + \sum_{b < |k| < 2b} \frac{2b - |k|}{b} \Gamma_k. \quad (20)$$

Adding (19) in the form $b\Lambda_0 = \sum_{|k| < b} (1 - |k|/b) \Gamma_k$, the coefficient of Γ_k becomes exactly 1 for $|k| \leq b$ and $(2b - |k|)/b \in (0, 1)$ for $b < |k| < 2b$, so

$$b\Lambda_0 + b(\Lambda_1 + \Lambda_1^\top) = S - \sum_{|k| \geq 2b} \Gamma_k - \sum_{b < |k| < 2b} \left(1 - \frac{2b - |k|}{b}\right) \Gamma_k = S + O(\rho^b), \quad (21)$$

using Lemma 11. This is the flat-top window of Politis and Romano [69] at block resolution; the point is that under geometric mixing its bias is exponentially small in b , whereas the batch-means bias in (i) is only $O(1/b)$. The algebraic identity $\widehat{S}^{\text{FT}} = 2\widehat{S}^{\text{BM}}(2b) - \widehat{S}^{\text{BM}}(b)$ (using disjoint pairs) is immediate from $\widehat{S}^{\text{BM}}(2b) = \widehat{S}^{\text{BM}}(b) + (b/T) \sum (\bar{g}_{2k-1} \bar{g}_{2k}^\top + \text{transpose})$.

(iii) *Variance.* $\widehat{S}^{\text{BM}}$ is an average of $T' \asymp N/b$ terms that are themselves weakly dependent with geometrically decaying block-level autocovariances (Lemma 11); Assumption 5 plus fourth moments give $\text{Var}[\widehat{S}_{jk}^{\text{BM}}] = O(1/T') = O(b/N)$, and the same for the lag-one term, hence for $\widehat{S}^{\text{FT}}$.

Drift. Write $\bar{g}_t(\theta_{t-1}) = \varepsilon_t + \Xi_t$ with $\Xi_t = \widehat{H}_t \delta_{t-1} + O(\|\delta_{t-1}\|^2)$. Then $\widehat{S}^{\text{FT}}$ built from $\bar{g}_t(\theta_{t-1})$ differs from the idealised version by terms bounded by $b T'^{-1} \sum_{t \in \mathcal{T}} \{\|\Xi_t\|^2 + 2\|\Xi_t\| \|\varepsilon_t\|\}$. Two facts are needed. First, on the retained window $t \asymp T$, Theorem 4 gives $\mathbb{E}\|\delta_{t-1}\|^2 = O(\log t/(bt))$. Second, $\widehat{H}_t$ is a b -sample estimate of a $d \times d$ matrix, so for δ independent of block t , $\mathbb{E}\|\widehat{H}_t \delta\|_{\text{op}}^2 = \delta^\top \mathbb{E}[\widehat{H}_t^2] \delta \leq (1 + Cd/b) \|H\delta\|_{\text{op}}^2$ under Assumption 3 (the d/b term is the fluctuation $\mathbb{E}[(\widehat{H}_t - H)M(\widehat{H}_t - H)]$, whose trace contributes a factor $\text{tr}(HM)/b$). Hence

$$b T'^{-1} \sum_{t \in \mathcal{T}} \mathbb{E}\|\Xi_t\|^2 = O\left((1 + d/b) T'^{-1} \sum_t \log t/t\right) = O((1 + d/b) b \log^2 N/N) = O((b + d) \log^2 N/N),$$

and the cross term is of smaller order by Cauchy–Schwarz. Writing $\tau_{b,N} = (b + d) \log^2 N/N$, this is dominated by the $\sqrt{b/N}$ standard deviation in (iii) whenever $b + d = o(\sqrt{N})$ up to logarithms; in the reported experiments $b + d \leq 340$ and $N \geq 2.5 \times 10^4$. Note that $\tau_{b,N}$ grows as b falls below d , which is visible in Figure 2(b) and is why we do not recommend $b \ll d$.

(iv) *Rate.* The entrywise bounds give $\|\widehat{S}^{\text{FT}} - S\|_{\text{op}} \leq \|\widehat{S}^{\text{FT}} - S\|_F \leq d \max_{jk} |\widehat{S}_{jk}^{\text{FT}} - S_{jk}|$, hence $\|\widehat{S}^{\text{FT}} - S\|_{\text{op}} = O_p(d\sqrt{b/N} + \rho^b + \tau_{b,N})$; the factor d is the price of the conversion and we do not try to remove it (a matrix-Bernstein argument under an explicit sub-exponential condition would give $\sqrt{db/N}$). Choosing $b = \lceil c \log N \rceil$ with $c > 2/\log(1/\rho)$ makes $\rho^b = O(N^{-2})$ and gives $O_p(\sqrt{\log N/N})$. For $\widehat{S}^{\text{BM}}$ the bias term $1/b$ replaces ρ^b and the trade-off $\sqrt{b/N} \asymp 1/b$ gives $b \asymp N^{1/3}$ and $O_p(N^{-1/3})$. Finally, $\bar{H}^{-1} \rightarrow_p H^{-1}$ (Theorem 2) and $\widehat{S}^{\text{FT}} \rightarrow_p S$ give $u_j^\top \widehat{S}^{\text{FT}} u_j \rightarrow_p V_{jj}$, and Theorem 5 with Slutsky gives the studentised limit. $\square$

A.9 Curvature schedules (Proposition 10)

Let $\widehat{H}^{\text{gap}} = n^{-1} \sum_{j \leq n} Y_{jm}$. By stationarity,

$$n \text{Var}[\widehat{H}^{\text{gap}}] = \Gamma_0^H + 2 \sum_{j \geq 1} \left(1 - \frac{j}{n}\right) \Gamma_{jm}^H = \Gamma_0^H + 2 \sum_{j \geq 1} \Gamma_{jm}^H + o(1) = \Gamma_0^H + O(\rho^m),$$

using $\|\Gamma_k^H\|_{\text{op}} \leq C\rho^k$ (Lemma 11 applied to Y , which has $\eta' > 2$ moments by Assumption 3). For Bernoulli thinning, let $Z_i \stackrel{iid}{\sim} \text{Bernoulli}(p)$ be independent of the data and let

$$\widehat{H}^{\text{Bern}} = \left(\sum_{i \leq N} Z_i \right)^{-1} \sum_{i \leq N} Z_i Y_i$$

be the estimator with the *realised* normaliser — which is what the algorithm computes, since W counts the updates actually made. Using a fixed normaliser pN instead would leave a term $p(1-p) \text{vec}(H) \text{vec}(H)^\top$ coming from $\mathbb{E}[Y_i] = H \neq 0$, which is $\Theta(1)$ and would swamp the $O(1/m)$ effect; the ratio form removes it, because $\widehat{H}^{\text{Bern}} - H = (\sum_i Z_i)^{-1} \sum_i Z_i (Y_i - H)$ *exactly*. Writing $n_Z = \sum_i Z_i$ with $\mathbb{E}n_Z = pN = n$ and $n_Z/n \rightarrow 1$ a.s., the delta method for a ratio gives $n \text{Var}[\widehat{H}^{\text{Bern}}] = n^{-1} \text{Var} \left[\sum_i Z_i (Y_i - H) \right] + o(1)$, and

$$\text{Var} \left[\sum_i Z_i (Y_i - H) \right] = \sum_i p \Gamma_0^H + \sum_{i \neq j} p^2 \Gamma_{i-j}^H = Np \Gamma_0^H + 2p^2 \sum_{k \geq 1} (N-k) \Gamma_k^H.$$

Dividing by $n = pN$ and using $p = 1/m$, $n \text{Var}[\widehat{H}^{\text{Bern}}] = \Gamma_0^H + 2p \sum_{k \geq 1} \Gamma_k^H + o(1) = \Gamma_0^H + m^{-1}(S^H - \Gamma_0^H) + o(1)$. $\square$

Remark 6. The contrast is that random thinning leaves a fixed fraction p of *all* lag pairs in the sum, whereas a deterministic gap removes every lag below m . For Gaussian AR(1) covariates and $\alpha \equiv 1$, Γ_k^H decays like ϕ^{2k} and $\kappa_H = (1 + \phi^2)/(1 - \phi^2)$.

B Experimental details

B.1 Data-generating processes and their population targets

Every simulated design pairs a dependent covariate process with a dependent error or latent process, for the reason given in Remark 1. In all four, $\theta^* = (-1, \dots, 1)$ evenly spaced in $\mathbb{R}^d$, $d = 20$, and the covariate covariance is the ill-conditioned $\Sigma = Q \text{diag}((i/d)^c) Q^\top$ of Boyer and Godichon-Baggioni [9] with Q a random orthogonal matrix, $c = 2$ (condition number 400) except for the logistic design where $c = 1$ (condition number 20) to keep the linear predictor in a sensible range. The population (H, Γ_0, S) is fixed once per design; Monte Carlo replicates vary only the realised path, which is checked in `tests/test_oracles.py`.

AR-hom (closed form). $x_t = \phi x_{t-1} + \sqrt{1 - \phi^2} \Sigma^{1/2} \eta_t$, $u_t = \psi u_{t-1} + \sqrt{1 - \psi^2} \sigma_u \varepsilon_t$, $y_t = x_t^\top \theta^* + u_t$, with η, ε standard normal and independent, $\phi = \psi = 0.7$, $\sigma_u = 1$. Then $H = \Sigma$, $\Gamma_k = \sigma_u^2 (\phi\psi)^{|k|} \Sigma$ and $S = \sigma_u^2 \Sigma (1 + \phi\psi)/(1 - \phi\psi)$, so $\kappa_j \equiv 2.922$.

AR-het (closed form). $x_t = A_\phi x_{t-1} + \Sigma_\eta^{1/2} \eta_t$ with $A_\phi = \text{diag}(\phi_1, \dots, \phi_d)$, ϕ_j evenly spaced on $[0, 0.9]$. The stationary covariance is the Hadamard product $\Sigma_x = \Sigma_\eta \circ [1/(1 - \phi_i \phi_j)]$, which is positive definite because $[1/(1 - \phi_i \phi_j)] = \sum_{k \geq 0} (\phi \phi^\top)^{\circ k}$ is a sum of Hadamard powers of a rank-one positive semidefinite matrix whose first d terms are linearly independent when the ϕ_j are distinct. With AR(1) errors, $\mathbb{E}[x_t x_{t+k}^\top] = \Sigma_x A_\phi^k$, $\Gamma_k = \sigma_u^2 \psi^k \Sigma_x A_\phi^k$ and $S = \sigma_u^2 (\Sigma_x + \Sigma_x G_\psi + G_\psi \Sigma_x)$ with $G_\psi = \psi A_\phi (I - \psi A_\phi)^{-1}$. Because the ϕ_j differ, κ_j varies across coordinates, so a scalar correction cannot pass this design.

Markov (closed form). s_t is a $K = d + 1$ state Markov chain with $\mathbb{P}(s_t = s_{t-1}) = 0.97$, uniform off-diagonal transitions and uniform stationary law; $x_t = V_{\cdot, s_t} + \Sigma_v^{1/2} v_t$, so covariates are a Markov-modulated Gaussian mixture—non-Gaussian marginals, discrete latent state, regime (non-linear) dependence. The errors are AR(1) with $\psi = 0.8$. Then $\Sigma_x = V \text{diag}(\pi) V^\top + \Sigma_v$, $\mathbb{E}[x_t x_{t+k}^\top] = V \text{diag}(\pi) P^k V^\top$ and $S = \sigma_u^2 (\Sigma_x + R + R^\top)$ with $R = V \text{diag}(\pi) \{\psi P (I - \psi P)^{-1}\} V^\top$.

The choice of centres matters and is worth recording, because the obvious construction produces a design that *looks* strongly dependent but has no inferential consequence. With randomly drawn centres and $K \ll d$, the regime component $V \text{diag}(\pi) V^\top$ has rank K and sits in high-curvature directions of Σ_x ; the sandwich $H^{-1} \Gamma_k H^{-1}$ is then negligible against $H^{-1} \Gamma_0 H^{-1}$, whose mass is in the *low*-curvature directions, and κ_j came out in $[1.00, 1.05]$ for every setting we tried. We therefore take $K = d + 1$ and the centres to be a regular simplex scaled by the target covariance: $V = \sqrt{K} \text{chol}(\Sigma) U$ with $U \in \mathbb{R}^{d \times K}$ an orthonormal basis of $\mathbf{1}^\perp$ in $\mathbb{R}^K$, which gives $V\pi = 0$ and $V \text{diag}(\pi) V^\top = \Sigma$ exactly. The regime component is then full rank and proportional to Σ , so $R_k = \lambda_2^k \Sigma$ with λ_2 the chain’s second eigenvalue and $\kappa_j \rightarrow (1 + \psi \lambda_2)/(1 - \psi \lambda_2)$ as $\Sigma_v \rightarrow 0$: at our settings, $\kappa_j = 7.55$ in every coordinate against the noise-free limit 7.88.

Logistic (simulated oracle). x_t is AR(1) with $\phi = 0.7$; the latent error is $e_t = \text{logit}(\Phi(v_t))$ with v_t a unit-variance AR(1) with $\psi = 0.8$, so e_t has an exact standard logistic marginal and $y_t = \mathbf{1}\{x_t^\top \theta^* + e_t > 0\}$ satisfies $\mathbb{P}(y_t = 1 \mid x_t) = \sigma(x_t^\top \theta^*)$ *exactly*. The working logistic likelihood is therefore correctly specified marginally— θ^* is the true parameter and $\Gamma_0 = H$ by the information equality—while the score is serially correlated through v . This is the marginal-model situation of Liang and Zeger [55]. H is computed to machine precision by Gauss–Hermite quadrature using the decomposition $x = (\Sigma_x \theta^* / \sigma_\ell^2) \ell + x_\perp$ with $\ell = x^\top \theta^*$ the linear predictor and $x_\perp \perp \ell$; S has no closed form and is obtained from eight independent streams of 10^6 observations with the score evaluated at the known θ^* and a Bartlett long-run covariance at lag 100 (the score autocovariance decays like $\psi^k = 0.8^k$, so the truncation error is below 10^{-9}). The resulting relative Monte Carlo error of the sandwich diagonal is reported in ??’s caption and is 7.5×10^{-3} ; it is the only non-exact population quantity in the paper.

B.2 Real streams: features, segments and the Protocol B correction

Features. Traffic: intercept, standardised temperature and cloud cover, two daily harmonics and one weekly harmonic, a holiday indicator propagated from the labelled midnight row to the whole calendar day, and a weekend indicator ($d = 11$). Rainfall is *excluded*: `rain_1h` is identically zero over stretches of several thousand consecutive hours, so $\log(1 + \text{rain})$ is a constant column on some contiguous segments, which makes the design there exactly rank deficient and the per-segment population targets undefined. Air quality: intercept, standardised temperature, pressure, dew point, $\log(1 + \text{rain})$ and wind speed, plus the same harmonics ($d = 12$). Feature construction was fixed before any response was looked at, and `exp6_real.py` asserts that every evaluation segment has full rank, reporting the worst per-segment condition number.

Segments and hold-out. The leading 15% of each stream is reserved for choosing the first-order baselines’ step-size constants and is excluded from every evaluation segment; the constants are chosen per stream, and the grid and whether the optimum is interior are recorded in `results/exp6_real.json`. Evaluation segments are the disjoint contiguous blocks that follow.

Protocol B’s nominal level. Let R be the number of evaluation segments, all of the same length, and let $\hat{\theta}_k$ be the estimate from segment k and $\hat{\theta}_{\text{full}}$ the full-stream M -estimator. If (i) the segment estimates are

approximately uncorrelated, (ii) they have a common variance V , and (iii) $\hat{\theta}_{\text{full}} \approx R^{-1} \sum_k \hat{\theta}_k$, then

$$\text{Var}(\hat{\theta}_k - \hat{\theta}_{\text{full}}) = \text{Var}(\hat{\theta}_k) - \text{Var}(\hat{\theta}_{\text{full}}) = V(1 - 1/R),$$

so an interval built to cover θ^* with probability $1 - \alpha$ covers $\hat{\theta}_{\text{full}}$ with probability $2\Phi(z_{\alpha/2}/\sqrt{1 - 1/R}) - 1$, which is what the header of ?? reports. Assumption (iii) requires the full-stream estimator to be (approximately) the equally weighted average of the segment estimators, which holds when the segments have equal length and similar $\widehat{H}$; (i) is the same approximation that makes segments usable as replicates at all. The correction is small (it raises the target from 0.950 to about 0.955 at $R = 8$) and we report the raw coverage alongside it.

B.3 Hyperparameters

Unless a sweep says otherwise: block length $b = 20$ ($= d$), gap $m = b$ (one curvature slot per block), no Bernoulli thinning ($p = 1$), log-weights off ($w' = 0$), ridge $c_t = 0.01$, $q_r = 0.4$, initial curvature scale $\lambda_0 = 10^{-6}$ (relative to the first block’s mean curvature magnitude, so it is dimensionless), warm-up $c_w = 50$ curvature updates per parameter, retained window dyadic (last half to last three quarters of blocks). ?? varies each of these.

The dependence-blind competitor “streaming Newton, i.i.d. theory” uses the base method’s schedule exactly: every observation is a candidate curvature slot ($m = 1$) with Bernoulli retention $p = 1/d$, and the interval uses the i.i.d. plug-in $\widehat{\Gamma}_0 = N^{-1} \sum_i s_i(\theta_{t-1}) s_i(\theta_{t-1})^\top$.

First-order baselines use step $\gamma_t = c t^{-3/4}$ (averaged SGD) or a constant c with AdaGrad scaling, with c chosen from a 19-point logarithmic grid on $[10^{-3}, 10^{1.5}]$ minimising squared error on eight *pilot* streams whose seeds are disjoint from the evaluation streams; the selected constants are recorded in `results/exp1_main.json`. They are additionally handed the *true* Hessian for their sandwich. Random scaling uses the two-sided 95% quantile 6.811 of the pivot $|W(1)|/(\int_0^1 (W(r) - rW(1))^2 dr)^{1/2}$, taken from Kiefer and Vogelsang [43] as tabulated by Lee et al. [50, Table 2]. Our own simulation of that quantile, using the exact Karhunen–Loève representation $\int_0^1 B^2 = \sum_{k \geq 1} Z_k^2 / (k\pi)^2$ with Z_k i.i.d. standard normal independent of $W(1)$, gives 6.746 from 4×10^6 draws (the sampler is validated by reproducing $\mathbb{E} \int_0^1 B^2 = 1/6$ and $\text{Var} \int_0^1 B^2 = 1/45$); we use the larger published value, which is generous to that baseline. We also flag a normalisation trap: several papers quote 5.32 as “the 95% critical value”, which is the *one*-sided quantile and would make the random-scaling intervals substantially too narrow.

The iterate-path comparator. Figure 3(a) compares against overlapping batch means computed from the iterate path, in the matrix form of the estimator used in the dependent-data literature: with T recorded iterates, window length b_n , window averages $\bar{\theta}_{b_n,t} = b_n^{-1} \sum_{k=t}^{t+b_n-1} \theta_k$ and overall average $\bar{\theta}$,

$$\widehat{\Sigma}^{\text{it}} = \frac{N}{T} \cdot \frac{b_n}{T - b_n + 1} \sum_t (\bar{\theta}_{b_n,t} - \bar{\theta})(\bar{\theta}_{b_n,t} - \bar{\theta})^\top, \quad b_n = \lceil T^{3/4} \rceil,$$

which estimates the covariance of $\sqrt{N}(\bar{\theta} - \theta^*)$, i.e. the same Σ our sandwich estimates. It uses no gradients and no curvature estimate. We compute it from our own recorded trajectory, with cumulative sums, so it costs $O(Td)$ and $O(Td)$ memory; it is a reference, not a streaming method, and we make no claim about the constants of any particular published recursion.

Table 1: Cost. Wall-clock milliseconds for one pass over $N = 200,000$ observations, single-threaded, median of 7 runs, and the empirical exponent s in time $\propto d^s$ fitted on the four largest d . The i.i.d. plug-in score covariance that the independent-data theory prescribes costs $O(d^2)$ per *observation*, so the dependence-blind interval is more expensive than our long-run interval, which costs $O(d^2)$ per *block*.

Asymptotic exponents are the same exact counts evaluated at $N = 10^8$, where the one-off $O(c_w d^3)$ warm-up is negligible and (7) reduces to $O(dN)$; at the N used here the warm-up inflates the exponent, which is why the measured value exceeds one.

method	$d=5$	$d=10$	$d=20$	$d=40$	$d=80$	$d=160$	s_{time}	s_{flop}
s_∞								
BGSN (long-run interval)	78.0	11.6	31.3	78.4	448.1	2697.5	2.18	1.56
1.00								
with the i.i.d. plug-in variance	43.4	79.6	241.6	632.2	2355.5	12448.9	1.90	1.91
1.86								
streaming Newton, Bernoulli $p = 1/d$	46.3	21.3	15.1	169.5	457.7	2677.6	2.38	1.56
—								
streaming Newton, every observation ($p = 1$)	77.0	292.8	840.9	2214.2	9754.7	46629.9	1.95	1.97
1.97								
ASGD (first order, random scaling)	8.8	5.2	32.3	81.3	168.3	478.2	1.27	—
—								
offline HAC, $L = 100$ (<i>not streaming</i>)	1568.3	2296.9	6894.0	21089.4	—	—	—	—
—								

B.4 Reproduction and compute

make all runs everything. All BLAS libraries are pinned to a single thread and parallelism is over Monte Carlo replicates (`experiments/_env.py` explains why: on the machine used, an unpinned tall-skinny $(d, N) \times (N, d)$ product with $N \approx 4 \times 10^4$, $d \approx 12$ took 18 s against 5 ms pinned). Total compute for every number in the paper is a few CPU-hours. The machine was shared and its load varied by more than an order of magnitude during the runs, so Appendix B.5 reports (i) exact machine-independent operation counts computed from the algorithm itself and (ii) wall-clock times measured with the repetitions *interleaved* across methods and summarised by the minimum over repetitions — round-robin ordering exposes every method to the same load pattern, and external load can only make a measurement slower, so the minimum is the robust summary. The operation counts are the authoritative statement; the times corroborate them.

B.5 Additional results